

PREDICTING DIABETES HOSPITAL READMISSIONS

A Machine Learning Approach to Improve Patient Care and Resource Allocation in Kenyan Healthcare System



Capstone Project Defense



Our Team

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Presentation Outline

01. Context & Problem

Understanding diabetes burden in Kenya and the challenge of hospital readmissions affecting patient outcomes.

02. Data Overview

Exploring the dataset from hospital records including patient demographics and clinical information.

03. Analysis & Insights

Key findings from data exploration showing patterns in readmission rates and risk factors.

04. Model & Impact

How machine learning predicts readmission risk and enables targeted interventions for better care.

Project Objectives

Our Mission

Develop a data-driven solution to predict diabetes readmission risk, enabling healthcare providers to implement targeted interventions and improve patient outcomes while optimizing hospital resources.



Identify Risk Factors

Analyze patient data to determine key predictors of 30-day readmission among diabetes patients.



Build Predictive Model

Develop accurate machine learning model that flags high-risk patients before discharge for intervention.



Enable Better Care

Provide actionable insights for clinicians to personalize treatment plans and reduce preventable readmissions.





KEY QUESTIONS



Clinical Risk Factors:

Which patient characteristics and clinical factors are most predictive of 30-day readmission for diabetic patients in Kenya?



Resource

Optimization:

How can we best allocate limited healthcare resources (beds, specialists, follow-up care) to maximize impact on readmission rates?



Model

Performance

Can we achieve sufficient predictive accuracy (especially recall) to justify implementation in resource-constrained settings?

Data Preparation & Data Understanding

Hospital Electronic Health Records

Data Source

Hospital electronic health records from diabetes patient admissions over 10-year period

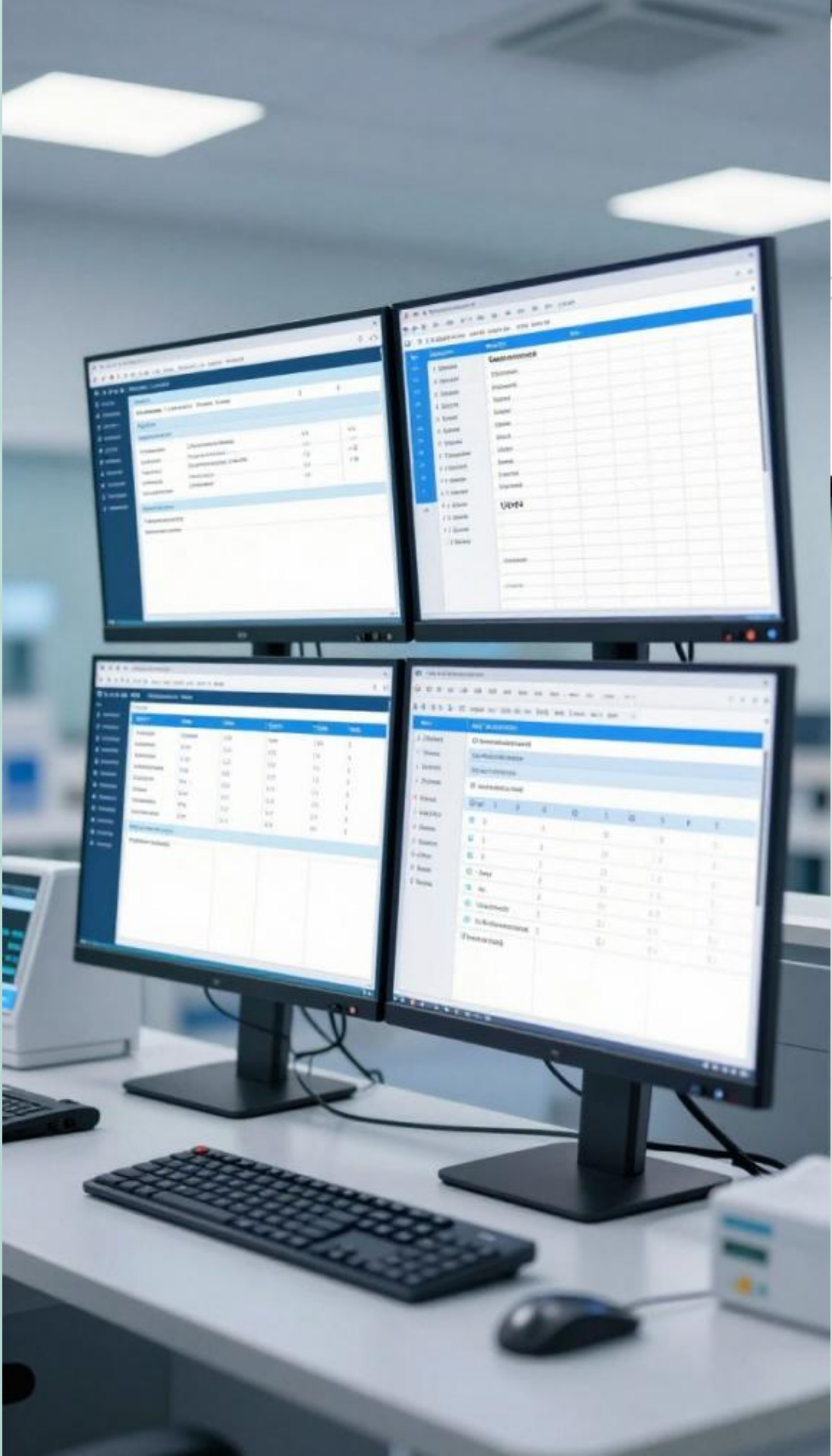
Dataset Size

100,000+ patient encounters across 130 hospitals

Key Features

- **Demographics:** Age, gender, race
- **Clinical:** Lab tests, diagnoses, procedures
- **Treatment:** Medications, insulin usage
- **History:** Previous admissions, emergency visits
- **Outcomes:** Length of stay, discharge status

All data anonymized to protect patient confidentiality

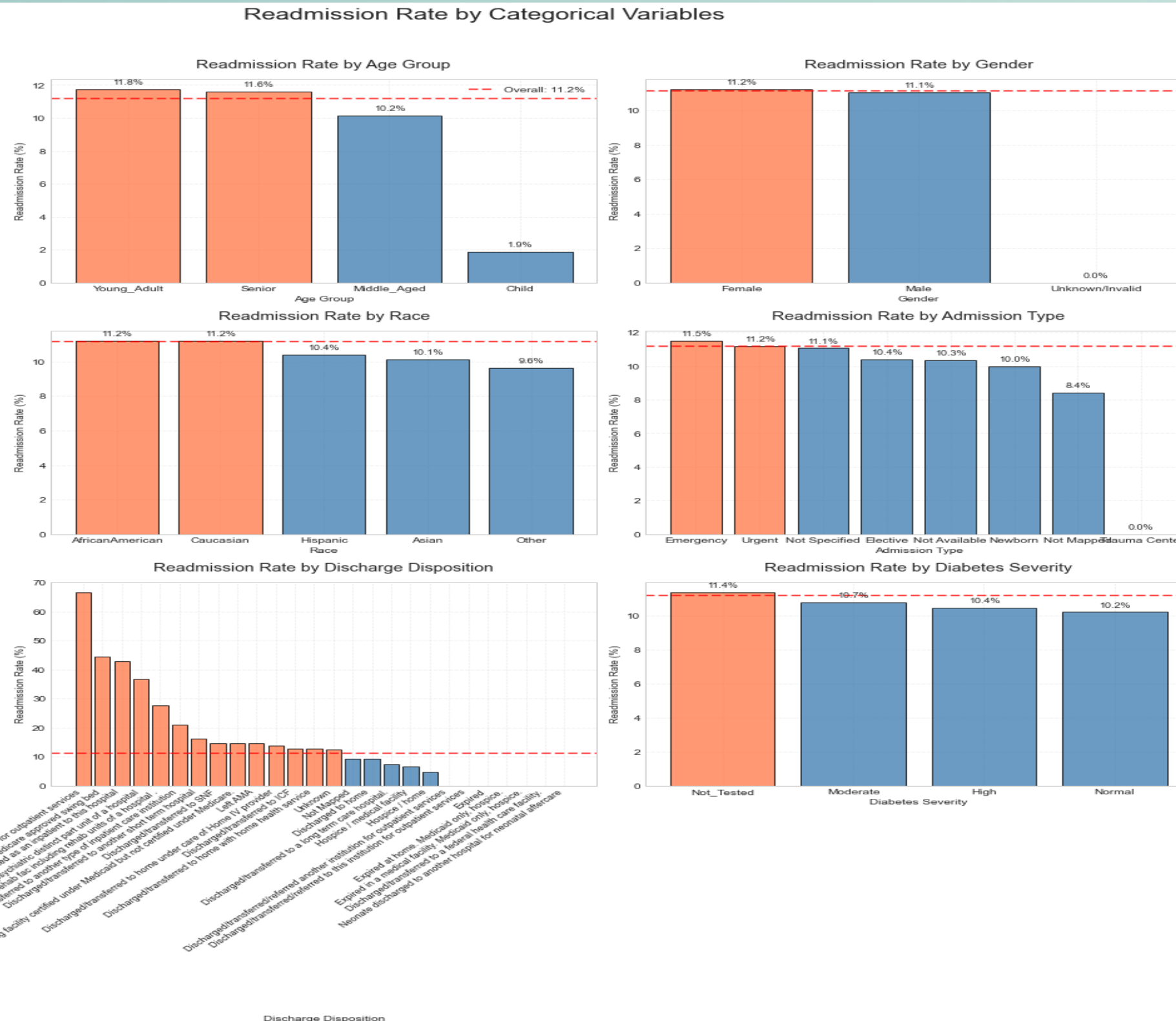


Readmission by Age Groups

Key Insights

Older patients show significantly higher readmission rates. Those aged 70+ have double the risk compared to younger patients, highlighting the need for age-specific discharge planning and follow-up care.

- **Age is a critical risk factor for readmission**
- **Elderly patients need enhanced post-discharge support**
- **Resource allocation should prioritize older demographics**



Hospital Stay Length vs Readmission

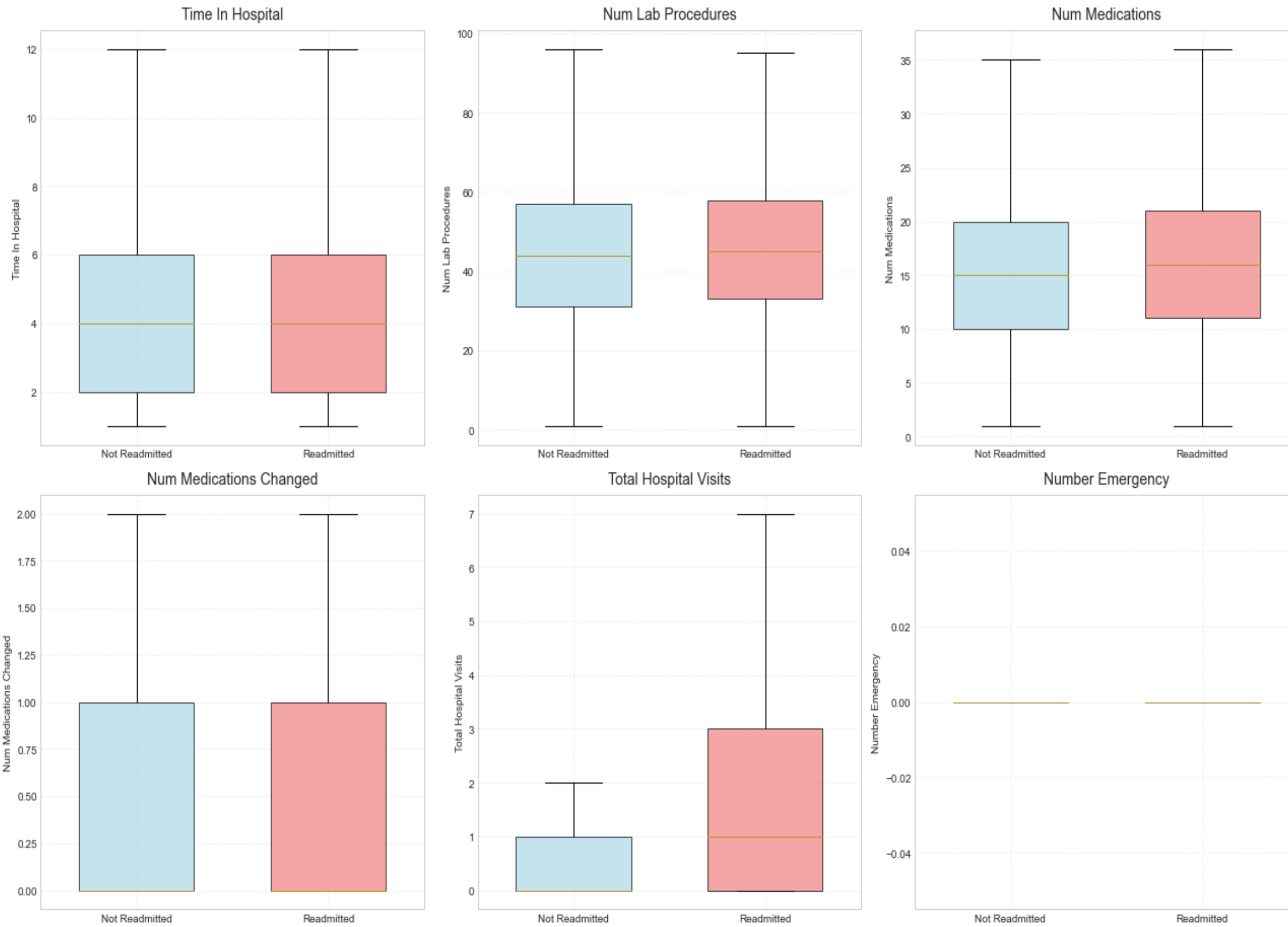
The Length of Stay Factor

Our analysis reveals an important finding: patients with very short hospital stays (1-2 days) have higher readmission rates. This suggests they may be discharged prematurely before stabilization. Conversely, stays beyond 7 days also show increased risk, indicating more severe cases requiring extended monitoring post-discharge.

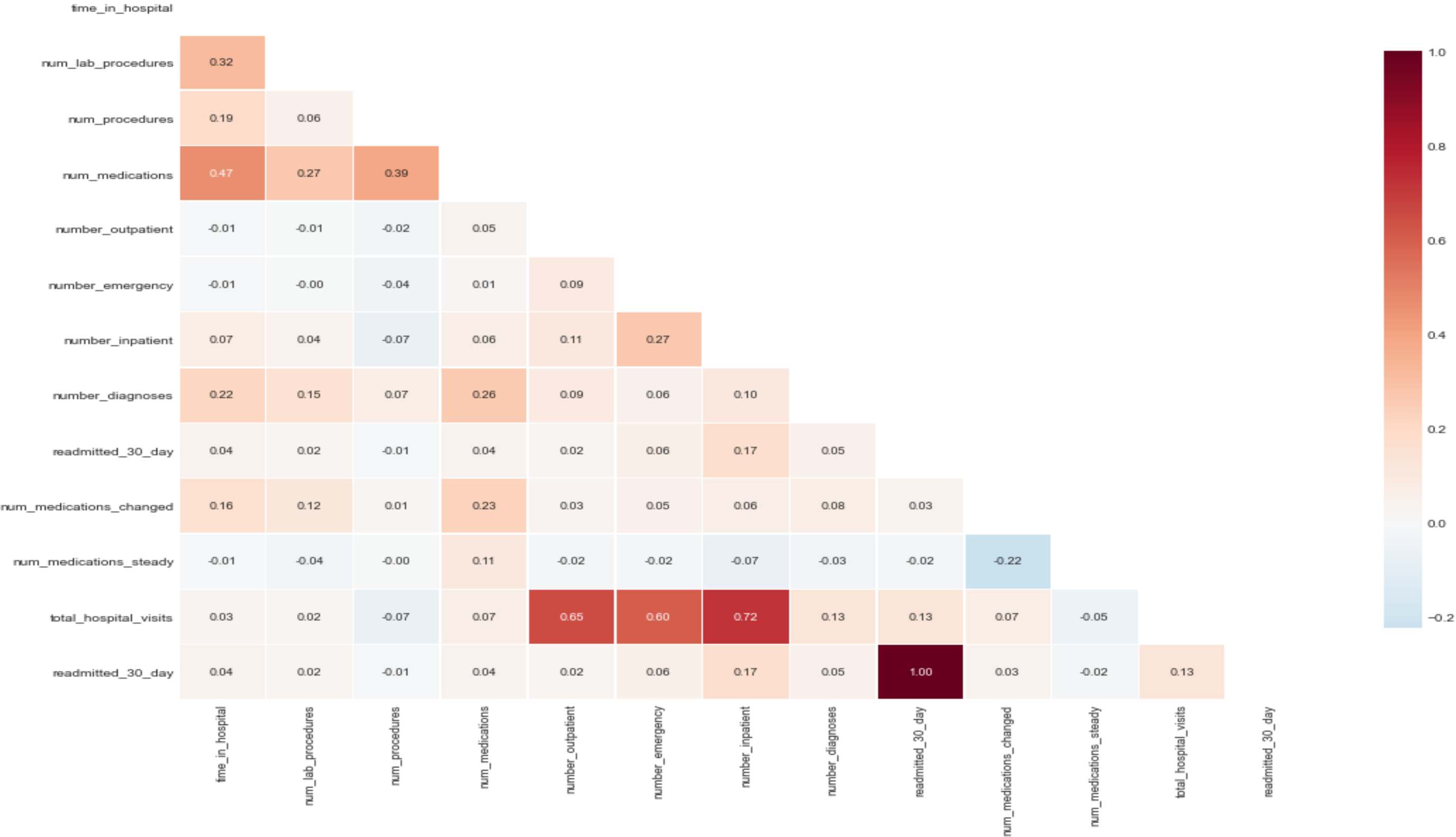
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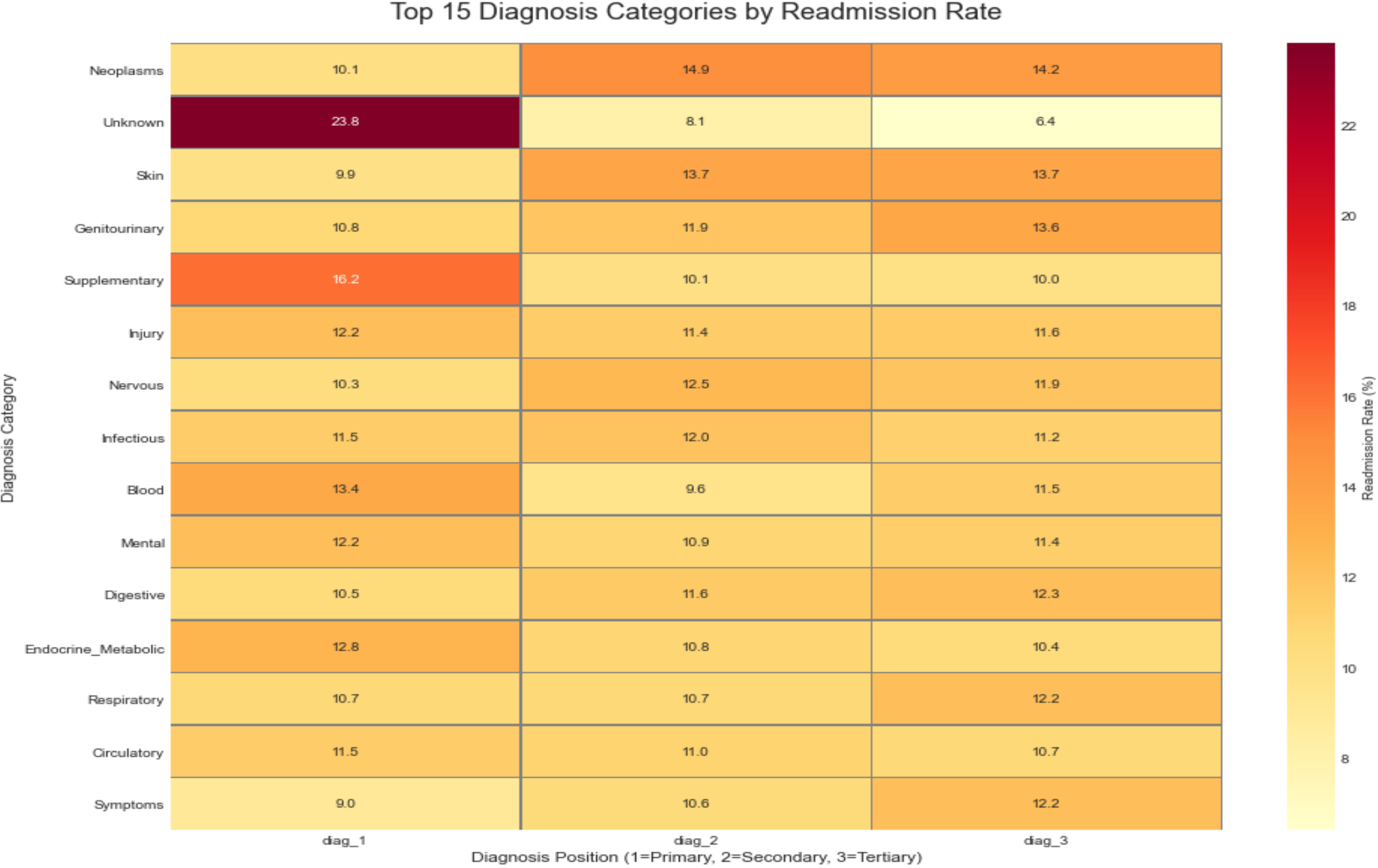
Optimal Stay Duration (days)

Distribution of Numerical Variables by Readmission Status



Correlation Matrix of Numerical Features





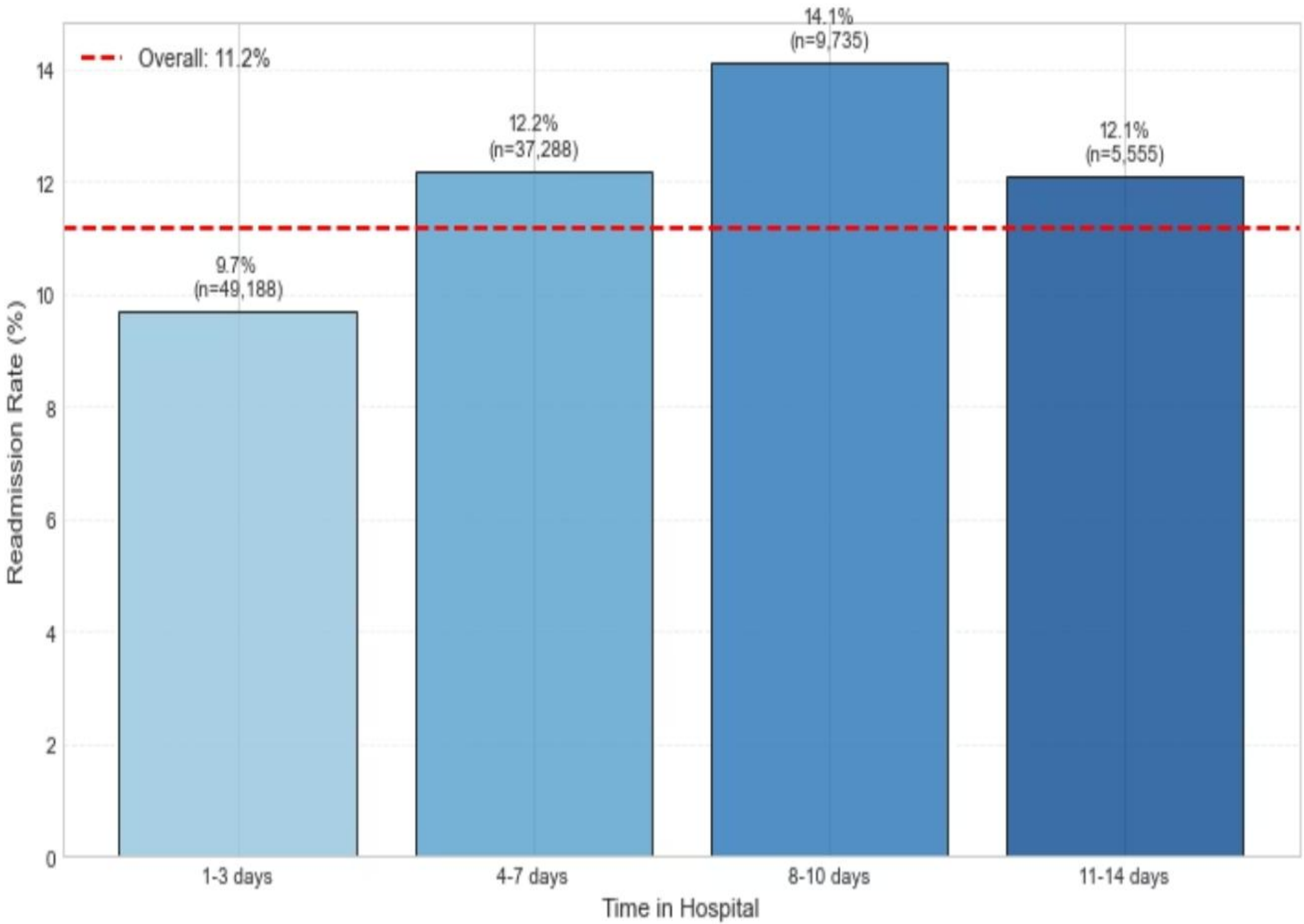
Impact of Comorbidities

Multiple Conditions Increase Risk

Patients with additional chronic conditions beyond diabetes face substantially higher readmission rates. Hypertension and heart disease are the most common comorbidities, present in over 60% of readmitted patients. This emphasizes the importance of comprehensive care that addresses all conditions simultaneously.

Patients with 3+ comorbidities have a 40% readmission rate compared to 15% for those with diabetes alone.

Readmission Rate by Length of Hospital Stay



Patient Demographics

Understanding Our Population

64

Average Patient Age
(Years)

52%

Male Patients

5.2

Average Hospital
Stay (Days)

16

Average Medications
Prescribed

Most patients are middle-aged to elderly adults (40–70 years), with slightly more males than females. The average hospital stay is about 5 days, and patients typically receive multiple medications reflecting the complexity of diabetes care and comorbid conditions.





Medication Impact on Outcomes

Treatment Complexity Matters

1-5 Medications

15% readmission rate – Simple treatment regimen

6-10 Medications

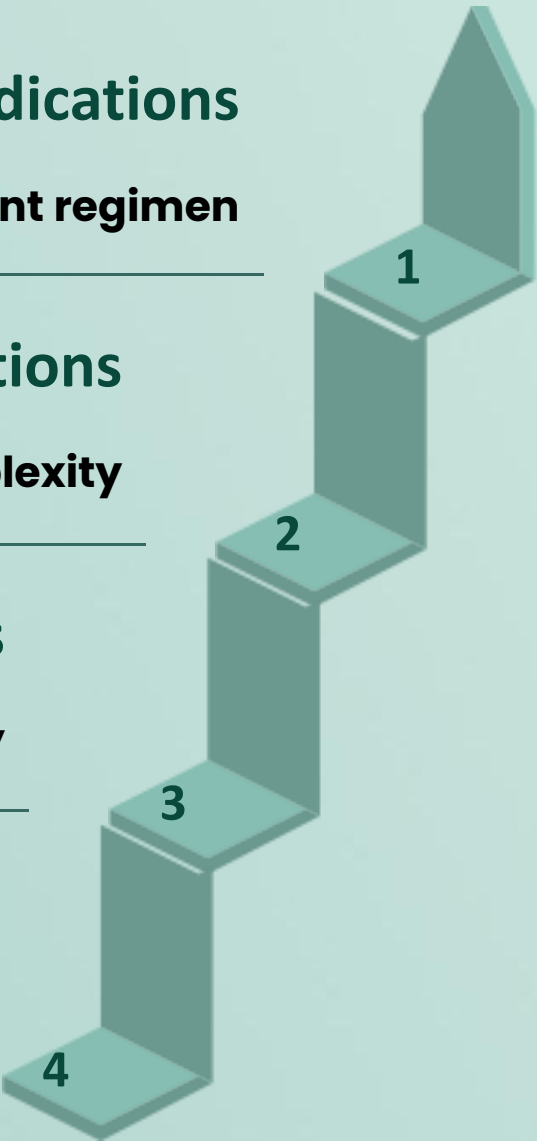
20% readmission rate – Moderate complexity

11-15 Medications

27% readmission rate – High complexity

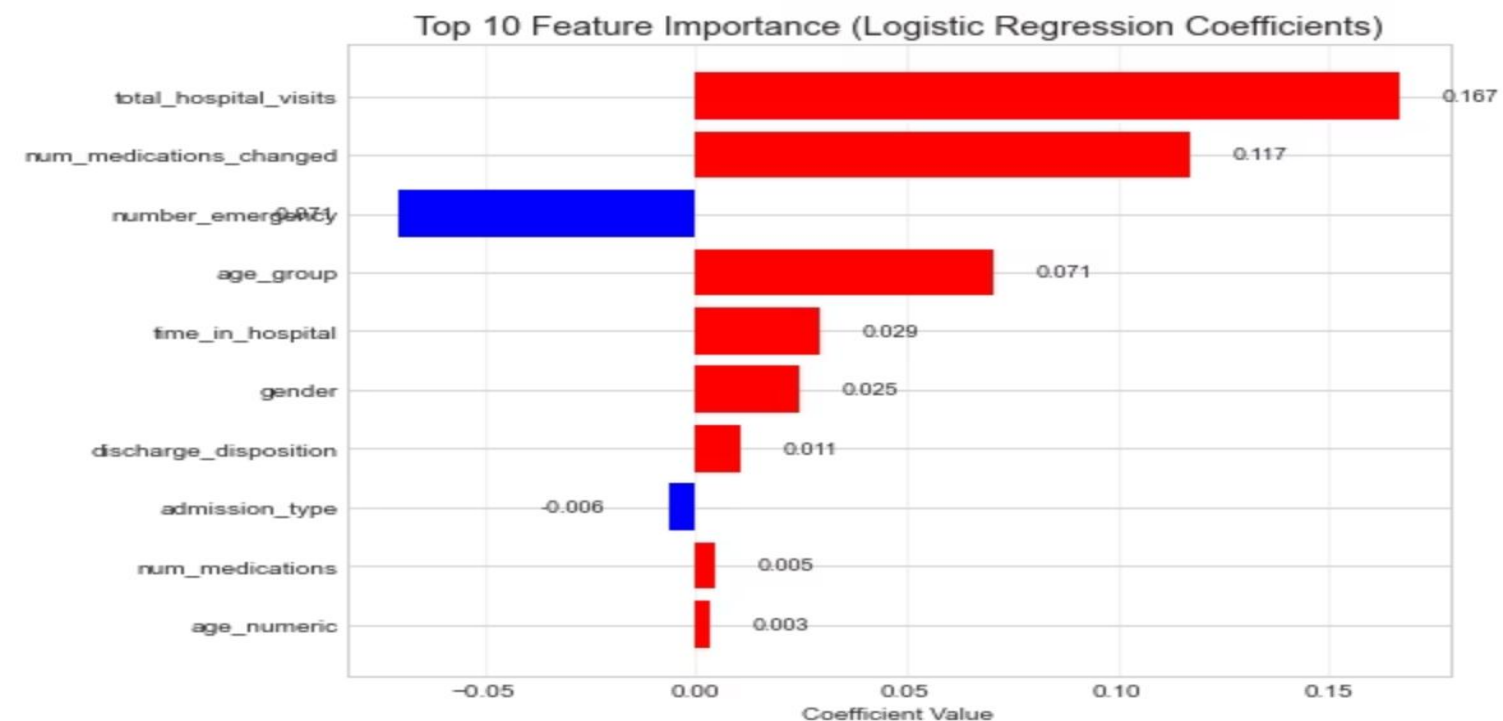
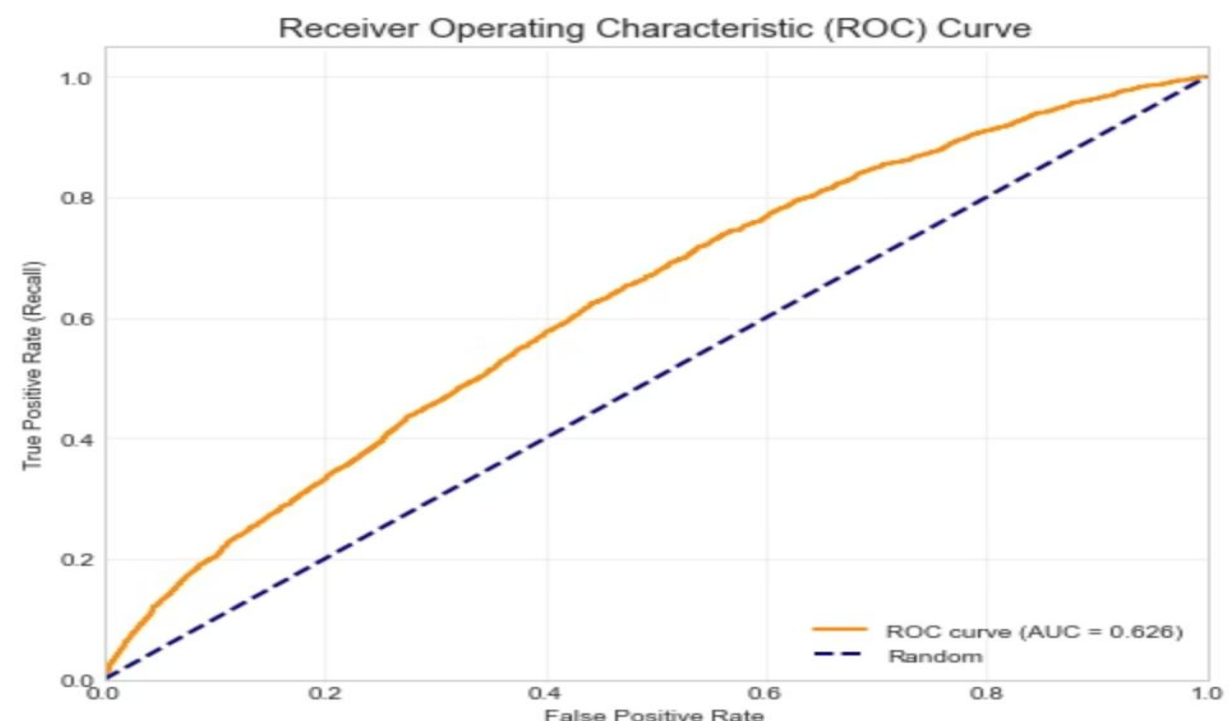
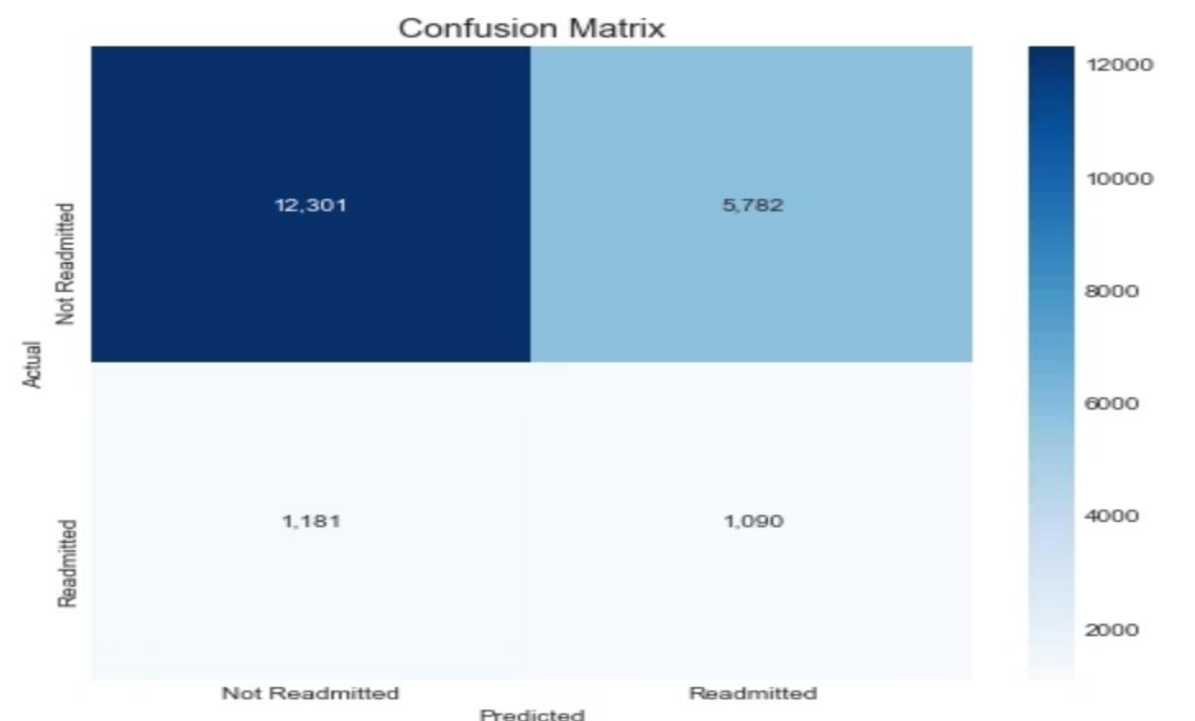
16+ Medications

35% readmission rate – Very high risk



Key Insight: Patients on complex medication regimens (16+ drugs) have more than double the readmission risk compared to those on simpler regimens. This indicates higher disease severity and potential medication adherence challenges.

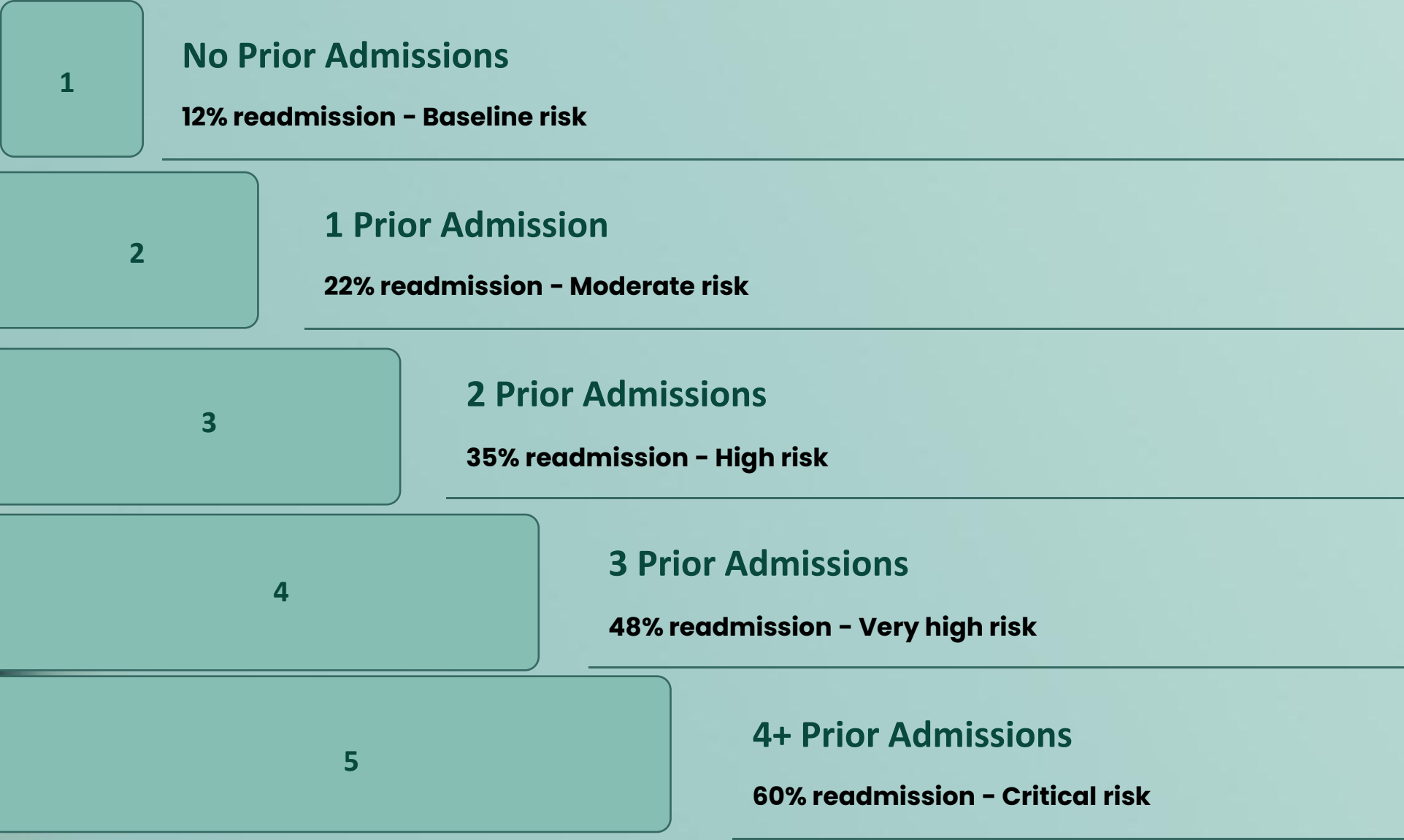
Logistic Regression Model Evaluation



This chart visualizes patterns in diabetes patient data related to healthcare resource utilization. Optimizing medication management and ensuring patient adherence are critical factors in reducing readmission rates and improving long-term diabetes management.

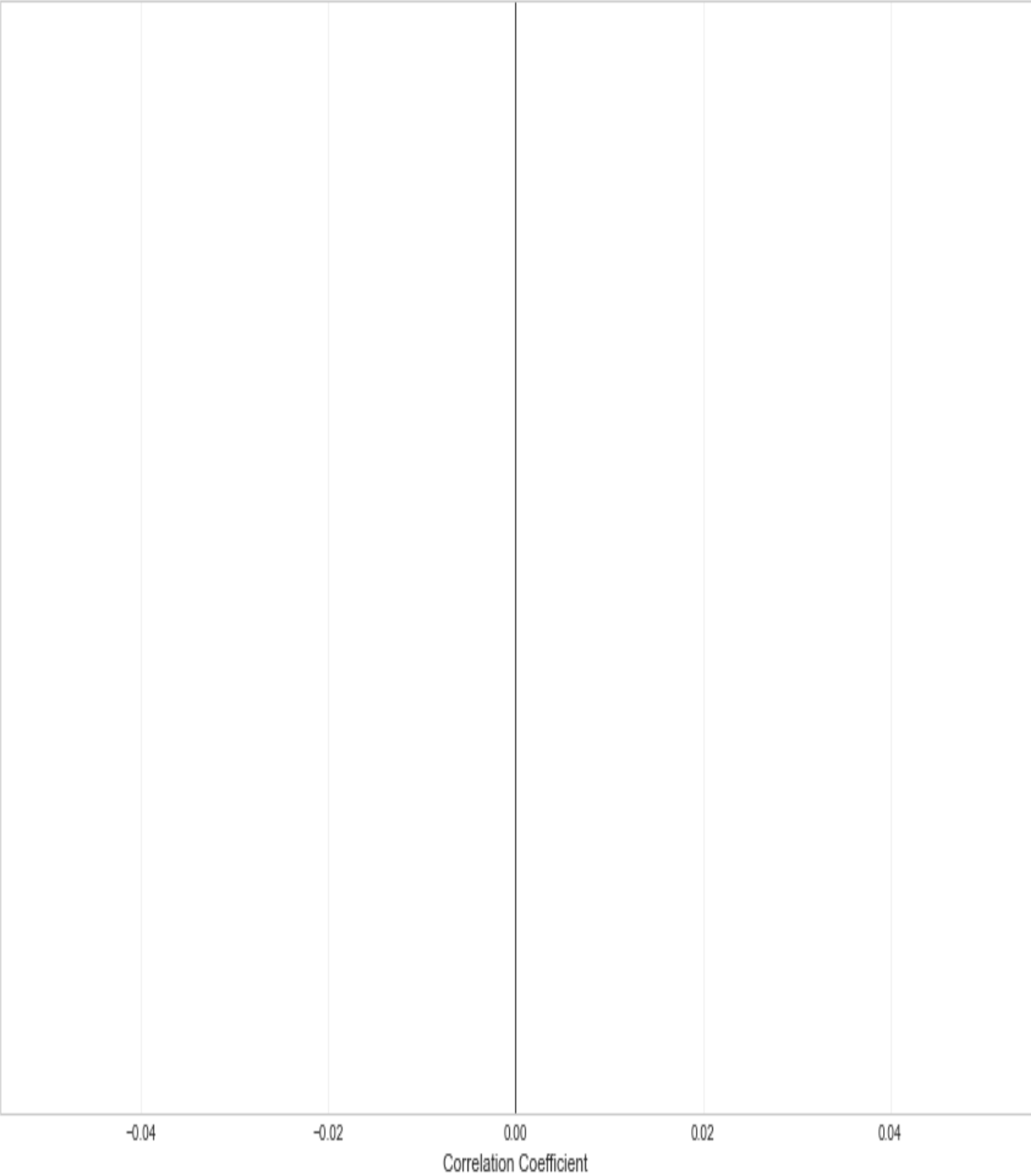
Previous Admissions - Strongest Predictor

Healthcare Utilization History



Number of previous inpatient visits is the #1 predictor in our model – patients with frequent admissions need intensive case management

Top Features Correlated with 30-Day Readmission



Machine Learning Model Performance

CATBoost Classifier Results

64.2%

Overall Accuracy

0.70

AUC Score

58.5%

Sensitivity (Recall)

64.1%

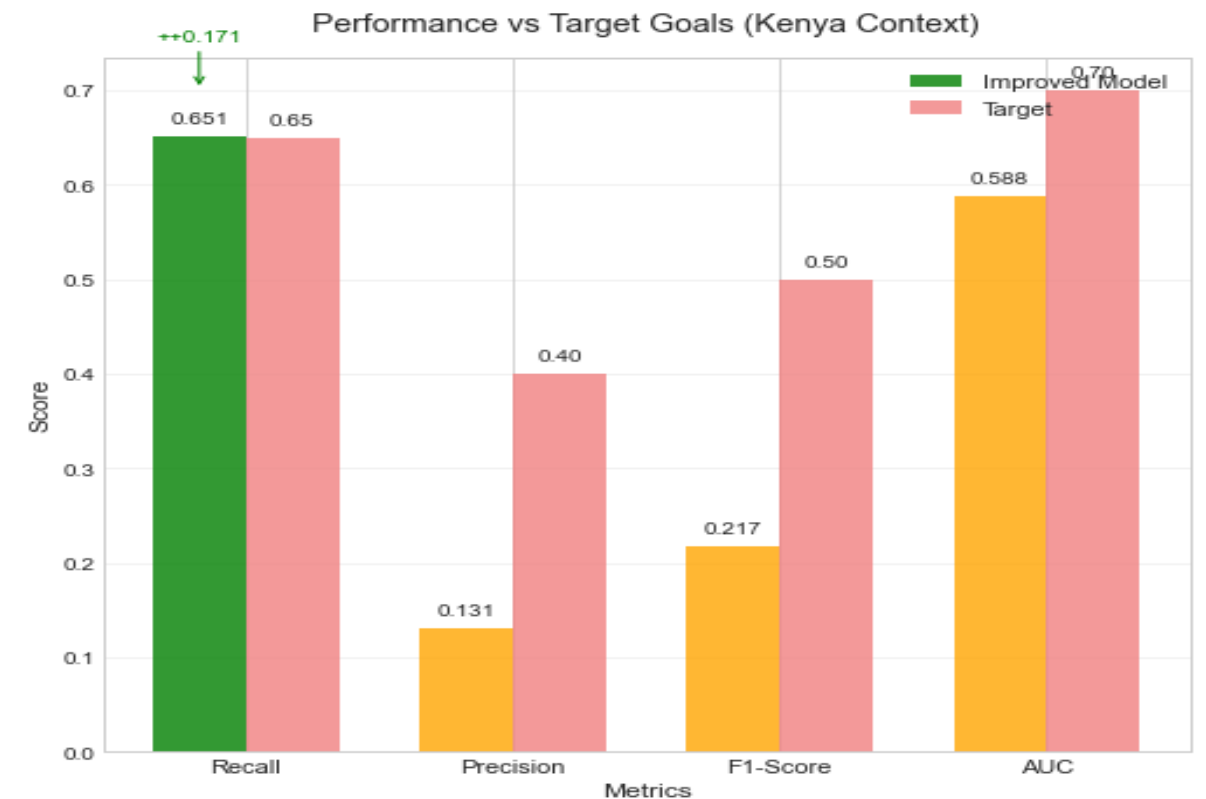
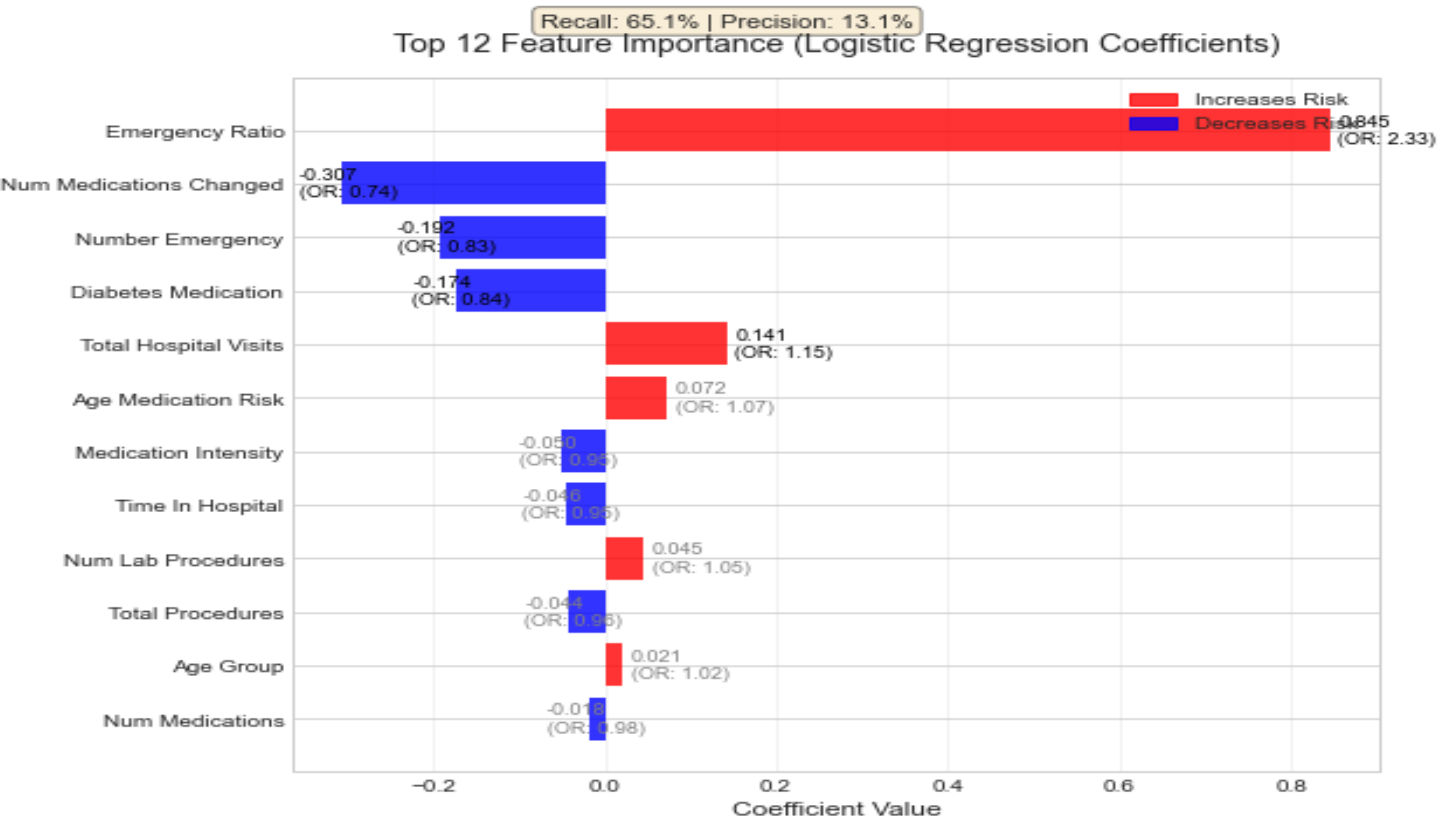
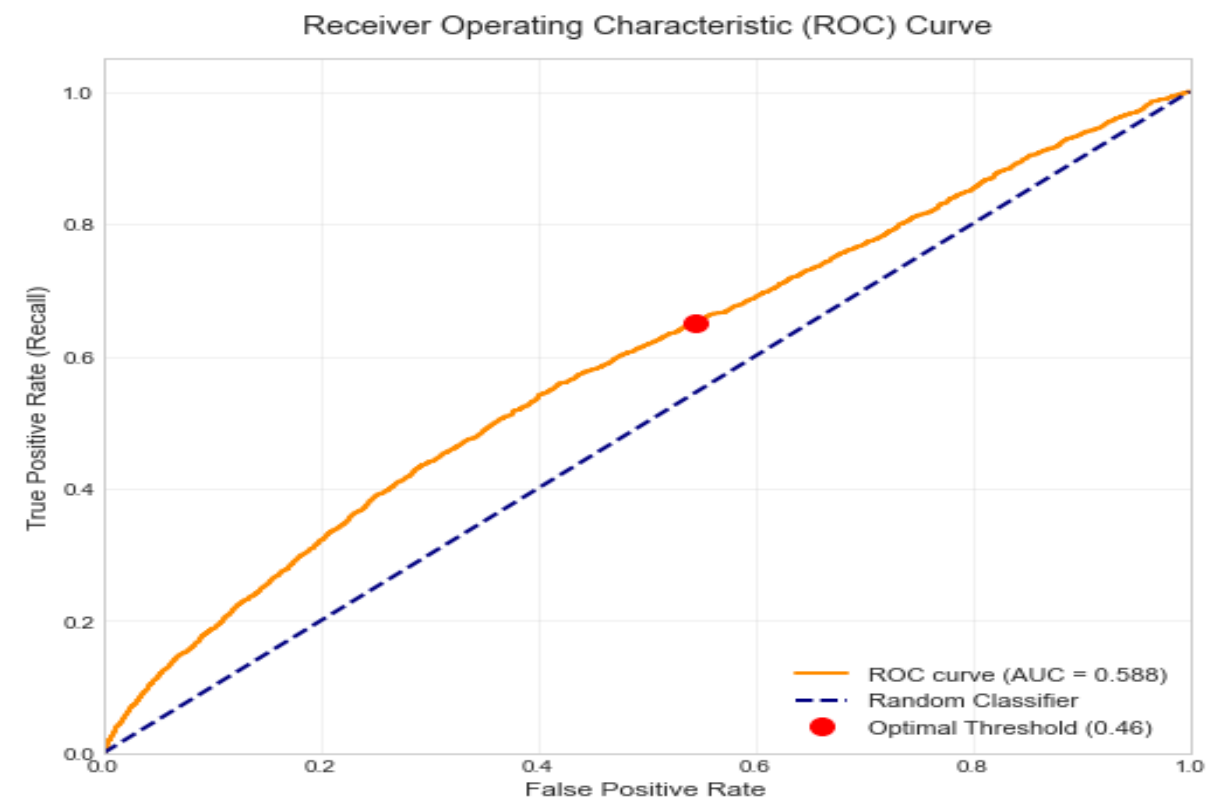
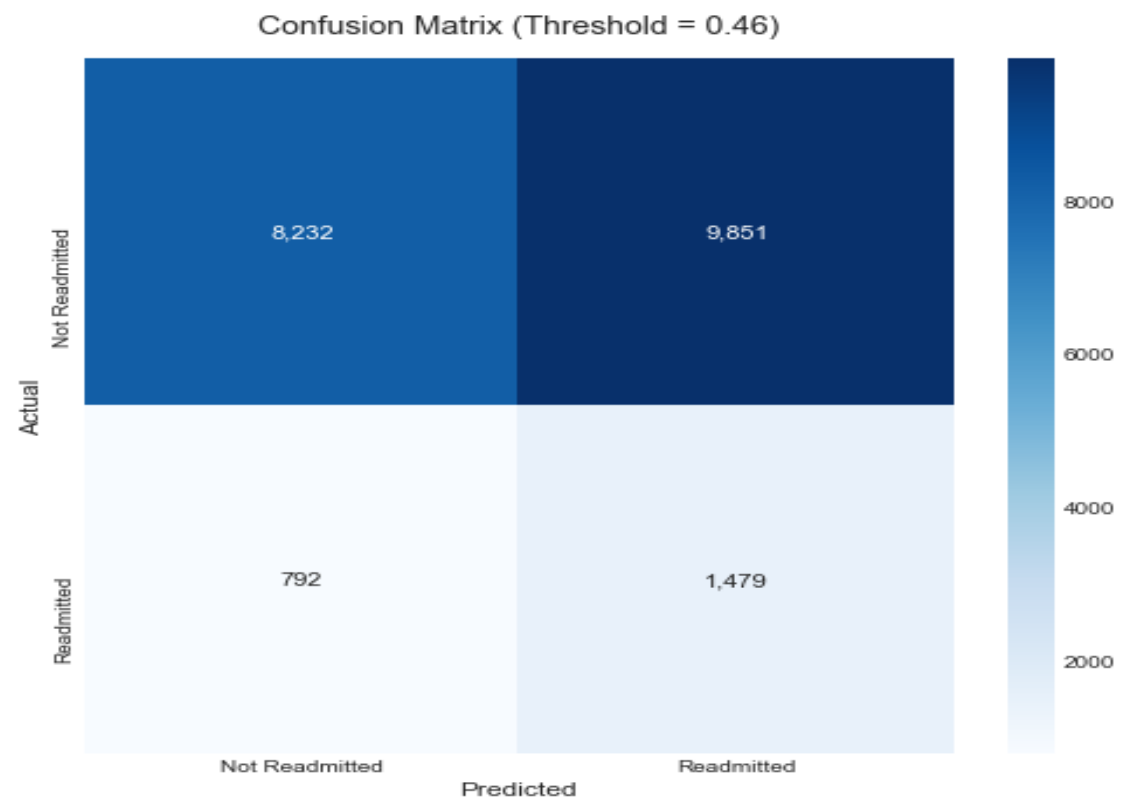
Specificity

What This Means: Our model correctly identifies nearly 2 out of 3 patients who will be readmitted, allowing healthcare providers to intervene early. The AUC score of 0.70 indicates good discriminatory power between high-risk and low-risk patients.

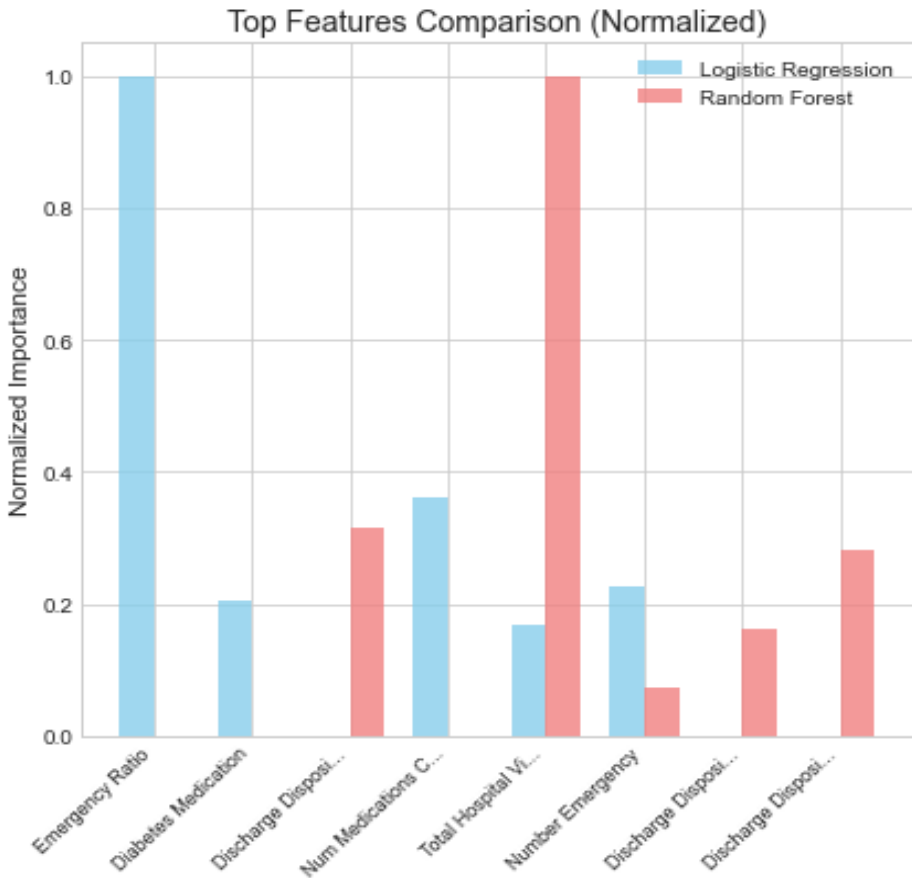
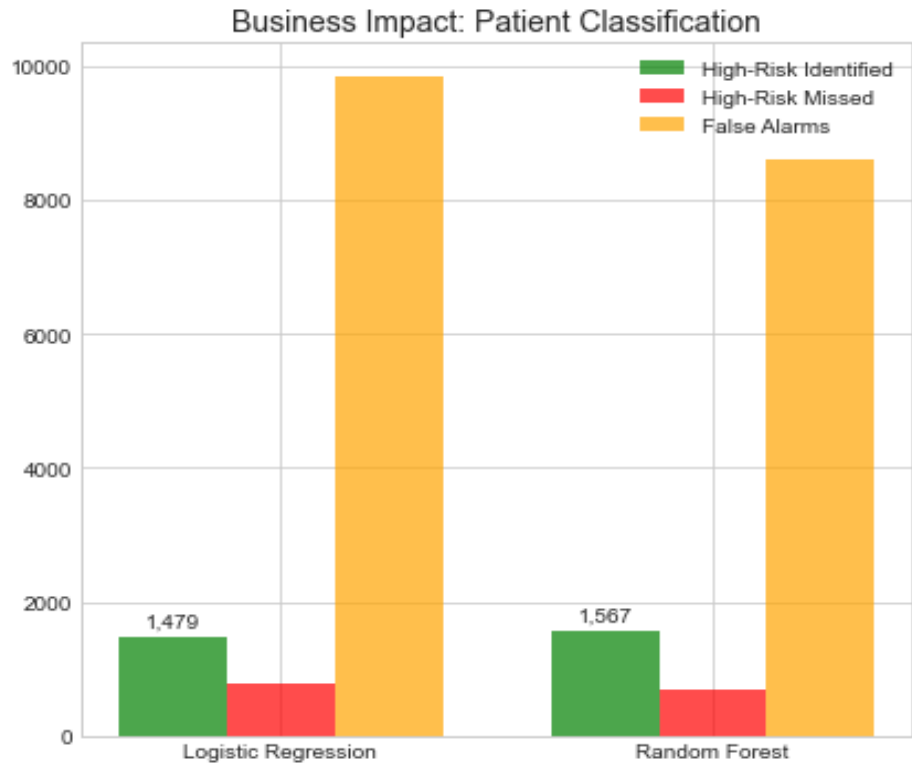
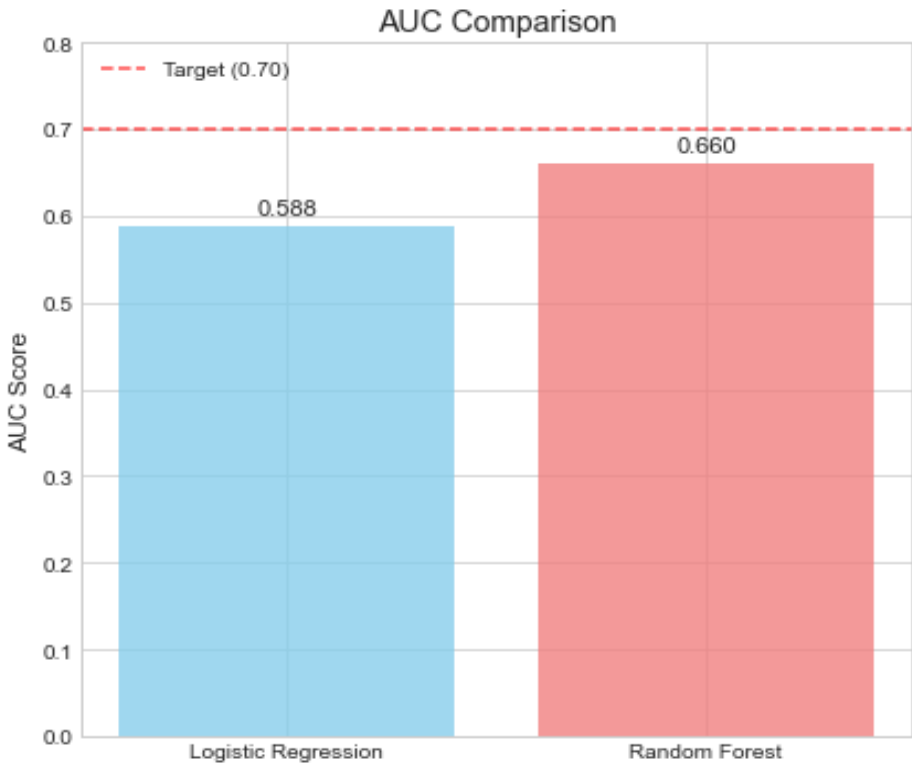
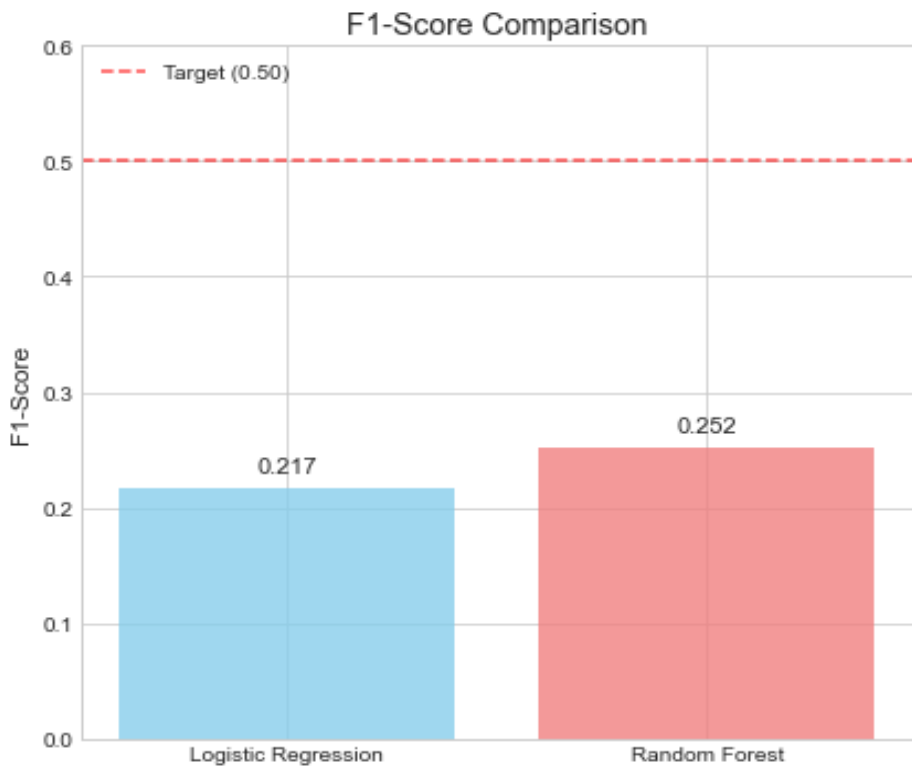
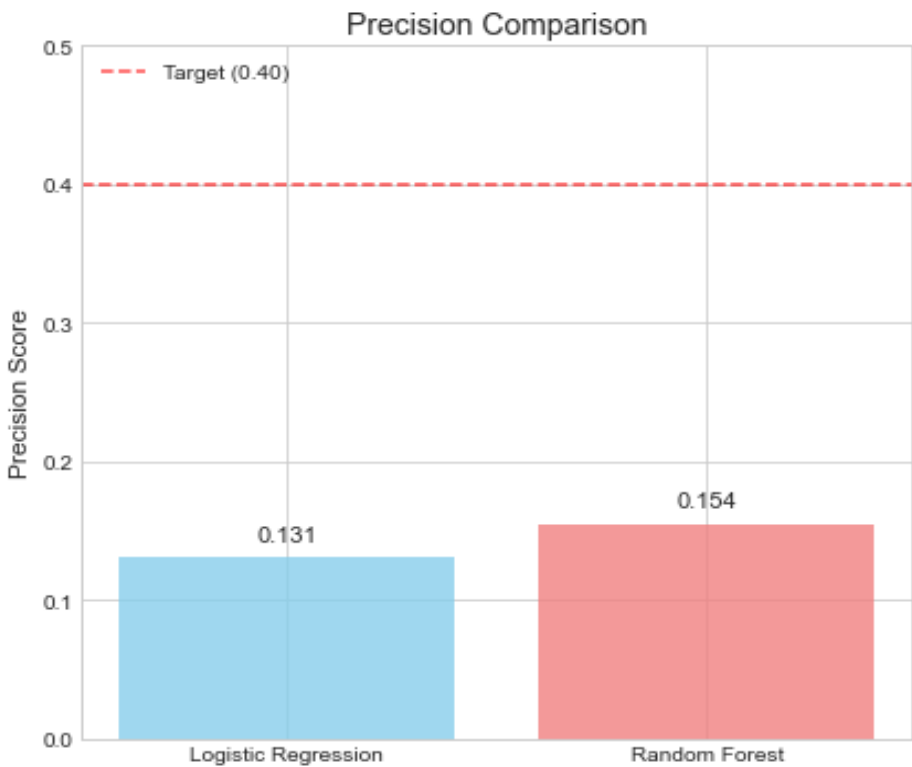
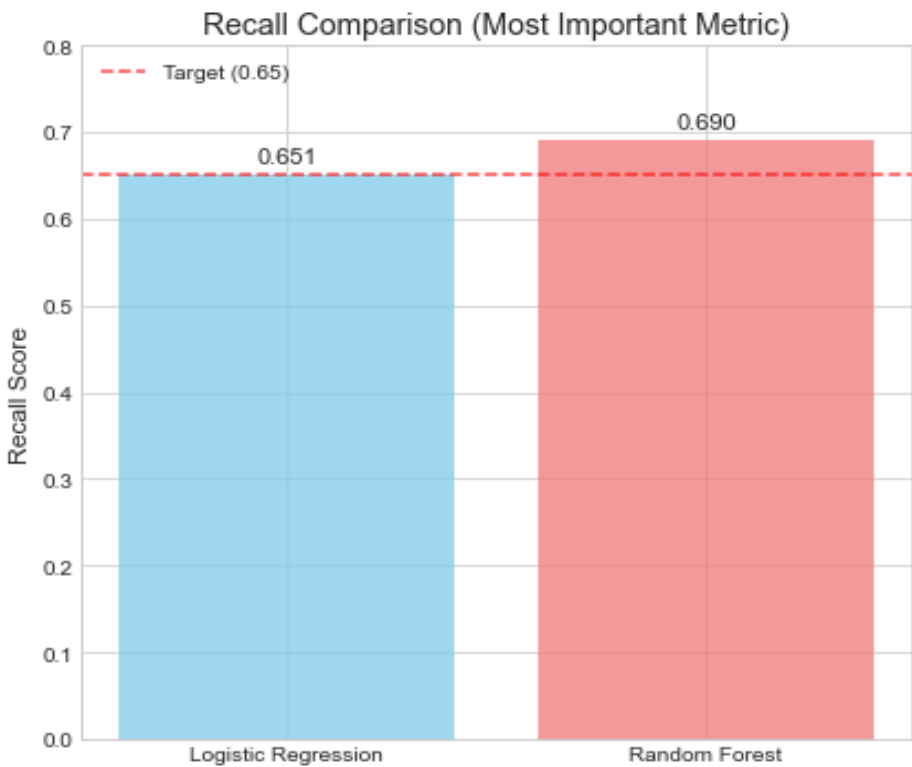


Machine Learning Model Performance

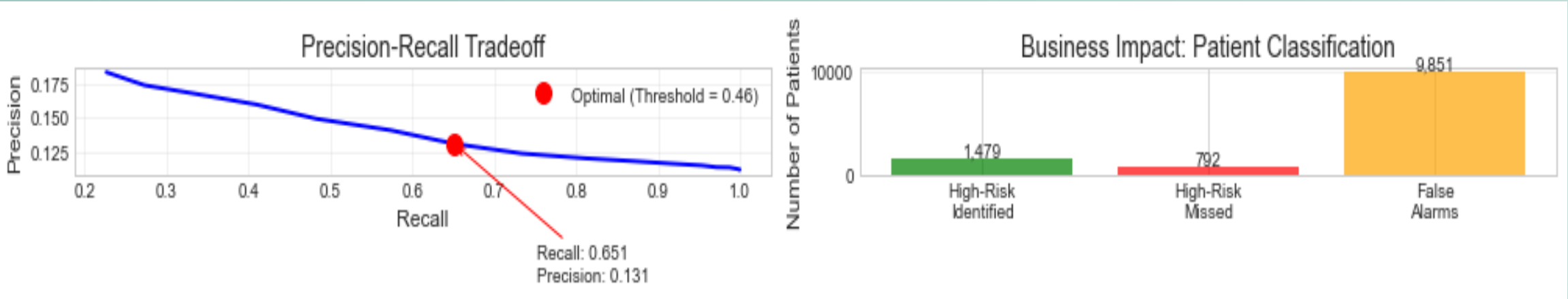
Improved Logistic Regression Model Evaluation



Model Comparison: Logistic Regression vs Random Forest



Top Risk Factors - Feature Importance



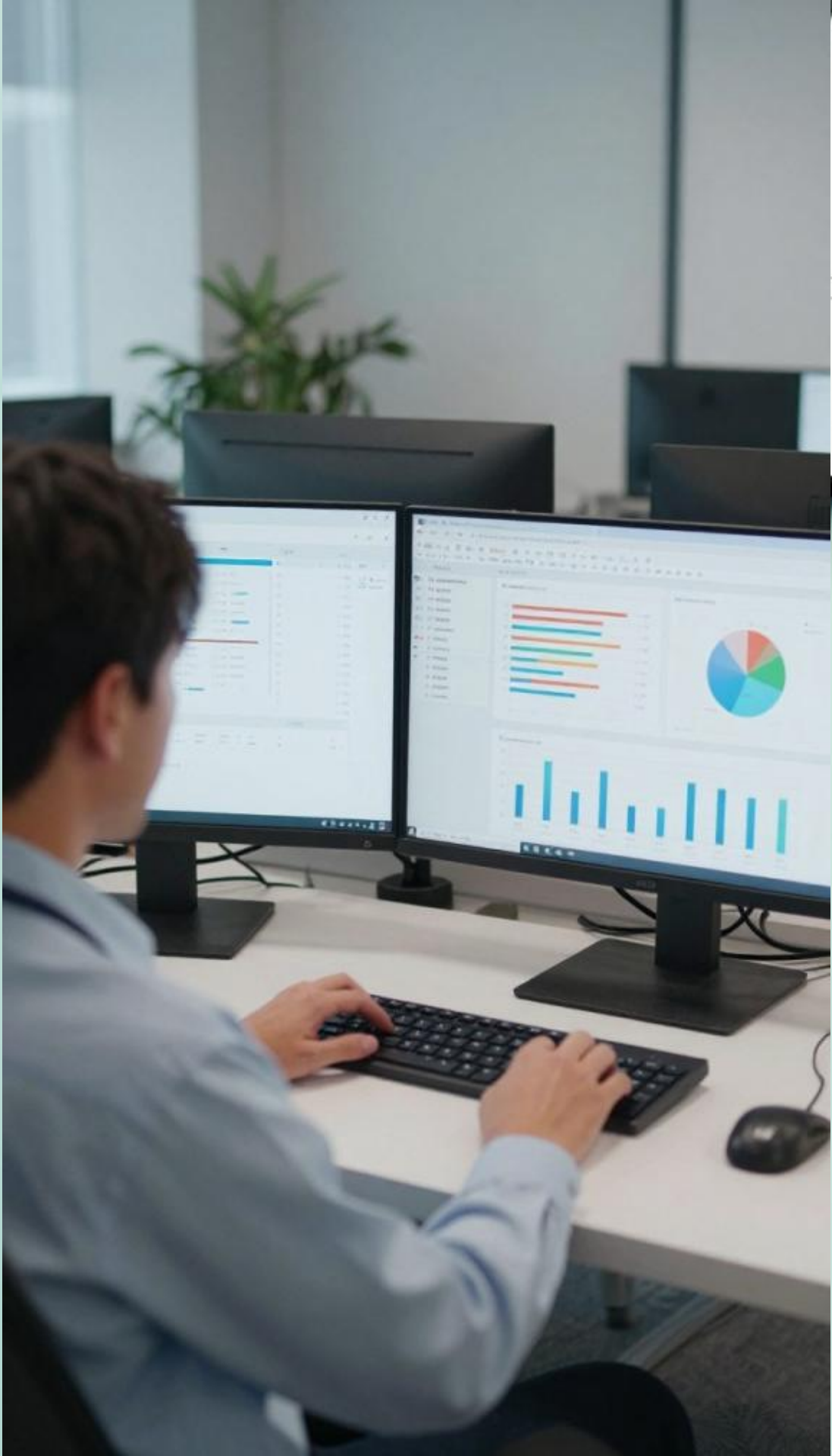
Healthcare Utilization

Previous inpatient and emergency visits dominate the model – frequent users of hospital services are at highest risk.



Treatment Complexity

Number of medications and lab procedures reflect disease severity and management difficulty – key predictors.



How the Model Works

Simplified Explanation for Non-Technical Audience

Input Data

Patient information: age, medications, diagnoses, previous admissions, lab results

Risk Calculation

For each new patient, model calculates readmission probability based on their specific profile

The model acts like an experienced doctor who has seen thousands of cases, but can process information instantly and flag risks that humans might miss.

Pattern Recognition

Model identifies patterns from 100,000+ past cases – learns which combinations predict readmission

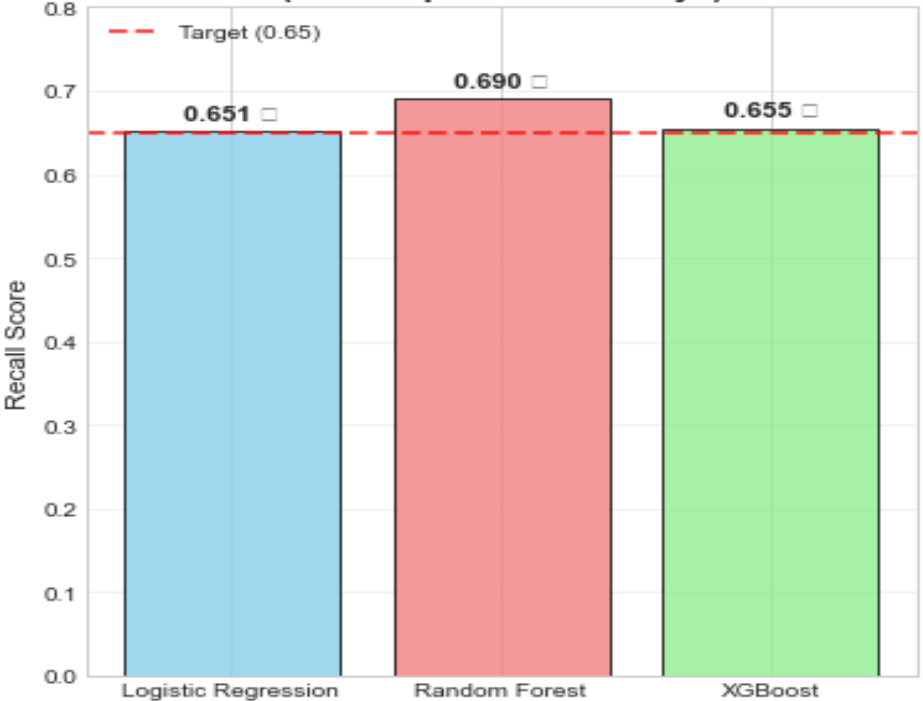
Risk Flag

High-risk patients flagged for clinician review and intervention before discharge

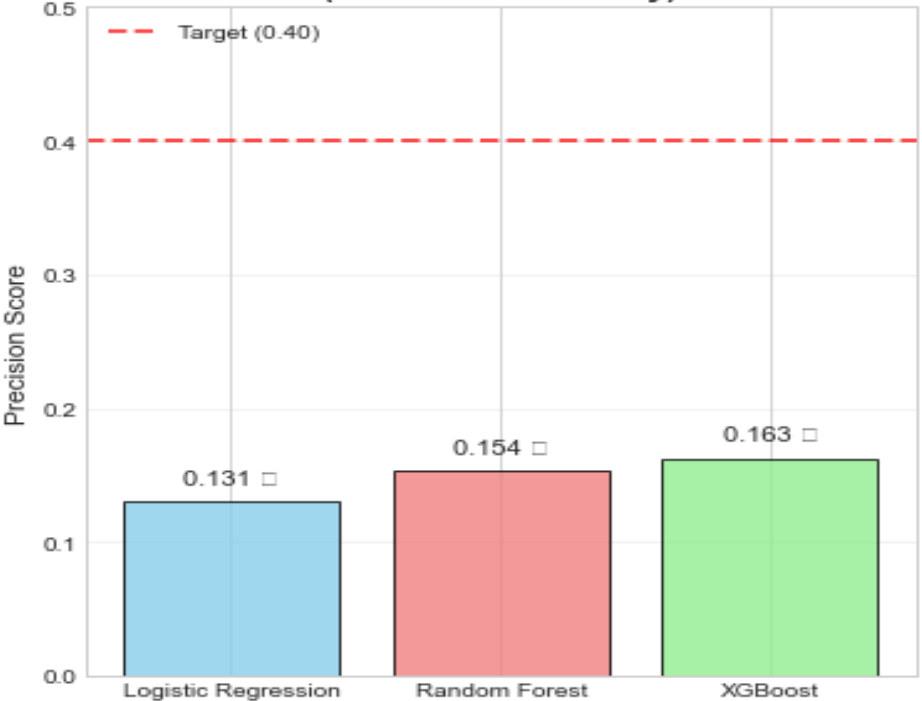


Model Comparison: Logistic Regression vs Random Forest vs XGBoost

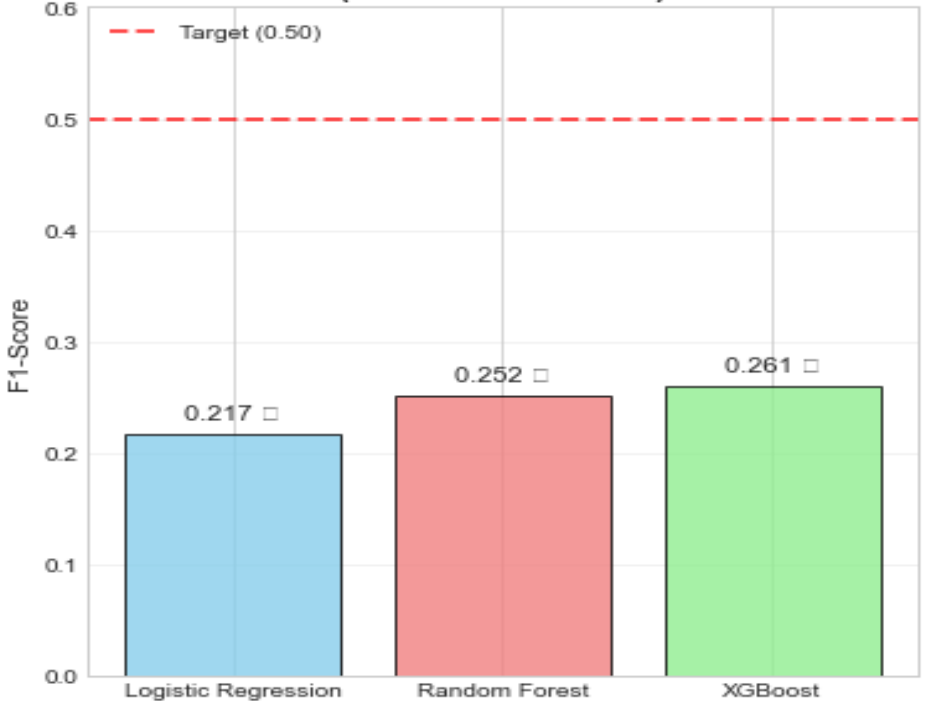
Recall Comparison
(Most Important for Kenya)



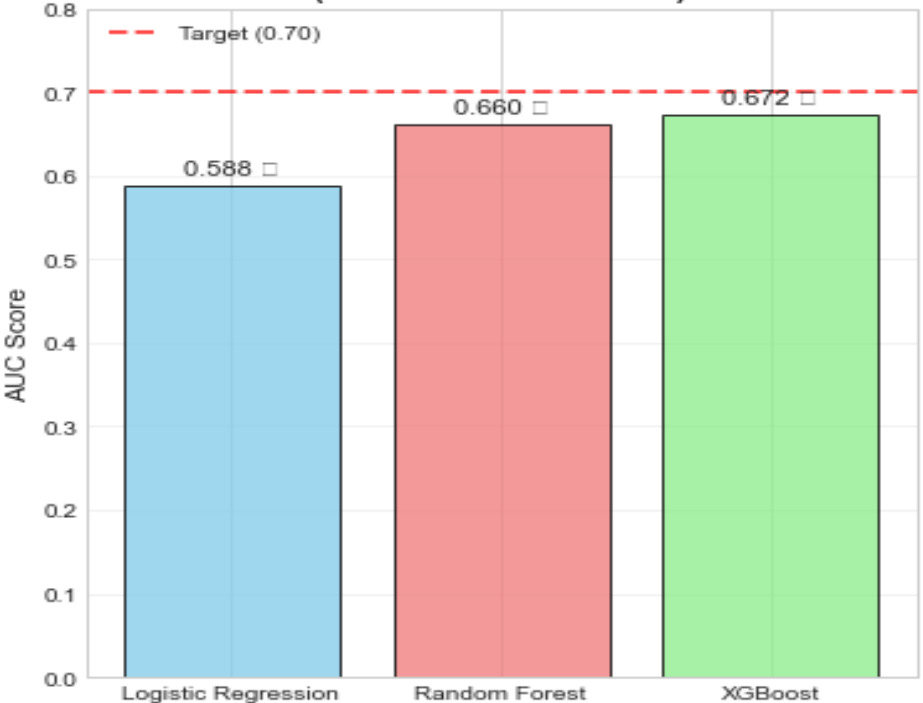
Precision Comparison
(Resource Efficiency)



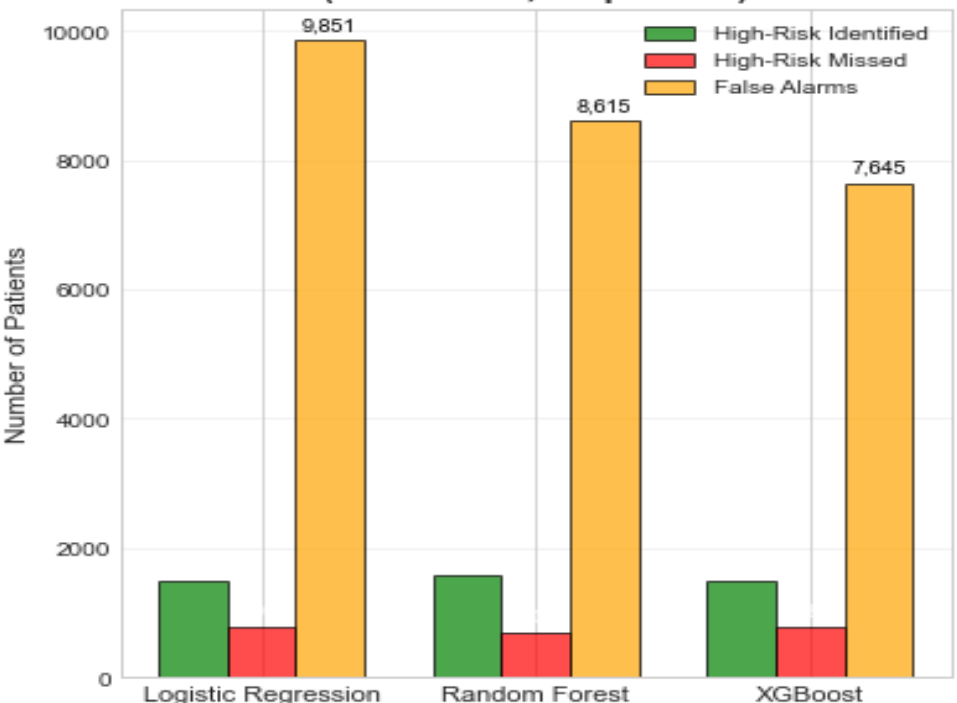
F1-Score Comparison
(Balanced Measure)



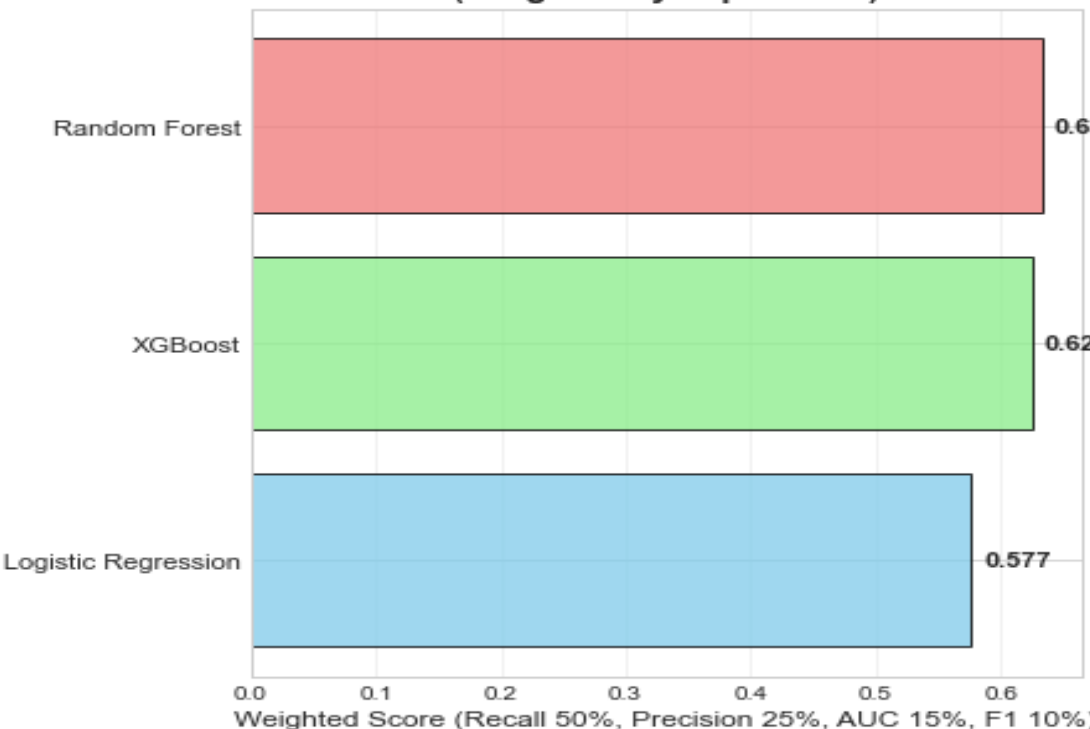
AUC Comparison
(Model Discrimination)



Business Impact: Patient Classification
(Test Set: 20,354 patients)



Model Ranking for Kenya Context
(Weighted by Importance)





Clinical Impact

Improving Patient Outcomes



Early Intervention

Identify high-risk patients before discharge and implement targeted care plans to prevent readmission



Personalized Care

Tailor treatment intensity, follow-up frequency, and support services based on individual risk levels



Enhanced Discharge Planning

Arrange home health visits, medication reviews, and patient education for those most likely to struggle



Remote Monitoring

Deploy telemonitoring and frequent check-ins for highest-risk patients to catch problems early

Operational Impact

Resource Optimization



Efficient Case Management

Allocate specialized case managers and resources to patients who need them most, rather than spreading thin across all patients



Capacity Planning

Predict bed demand more accurately by forecasting readmissions, enabling better staffing and resource allocation



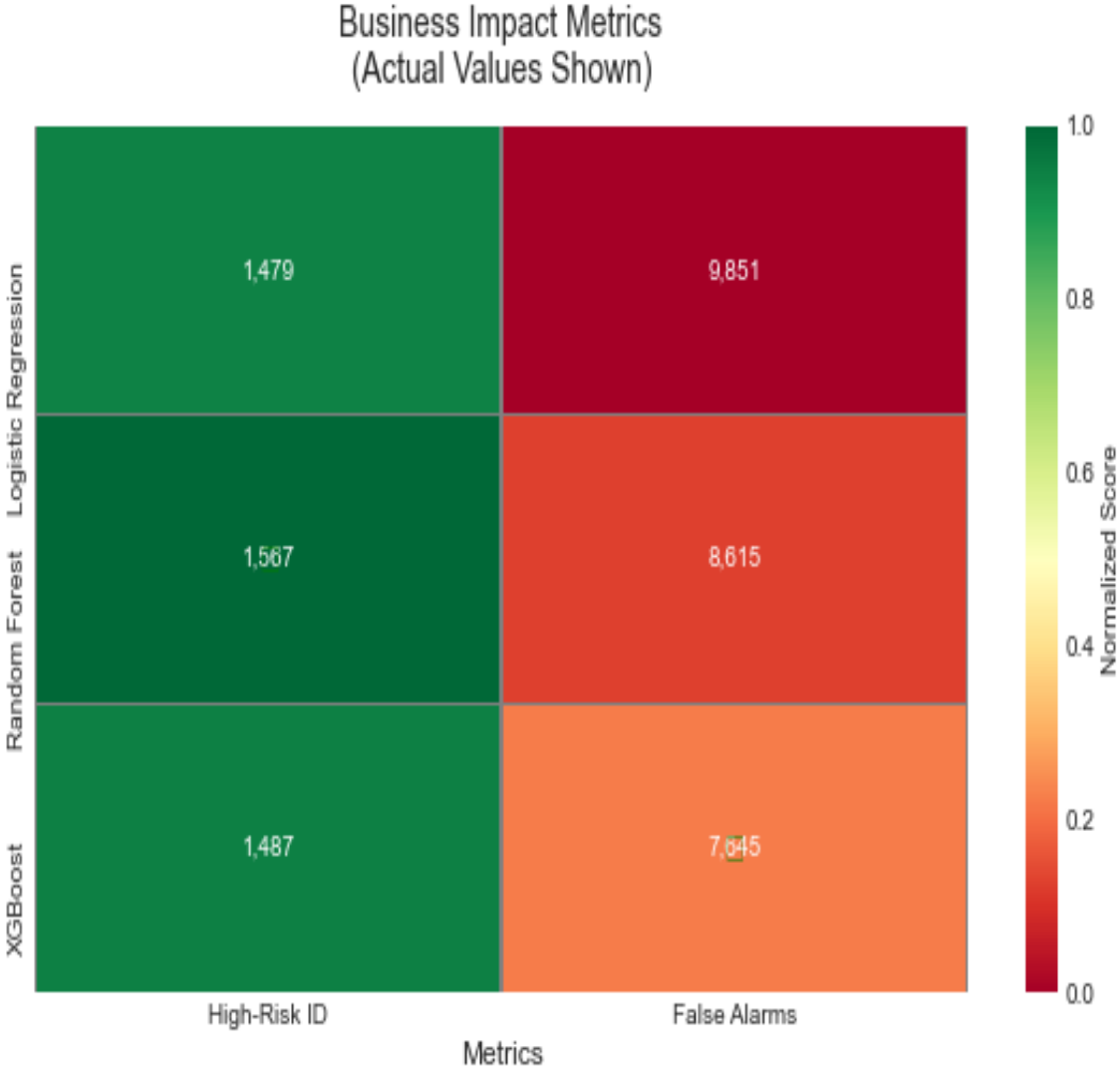
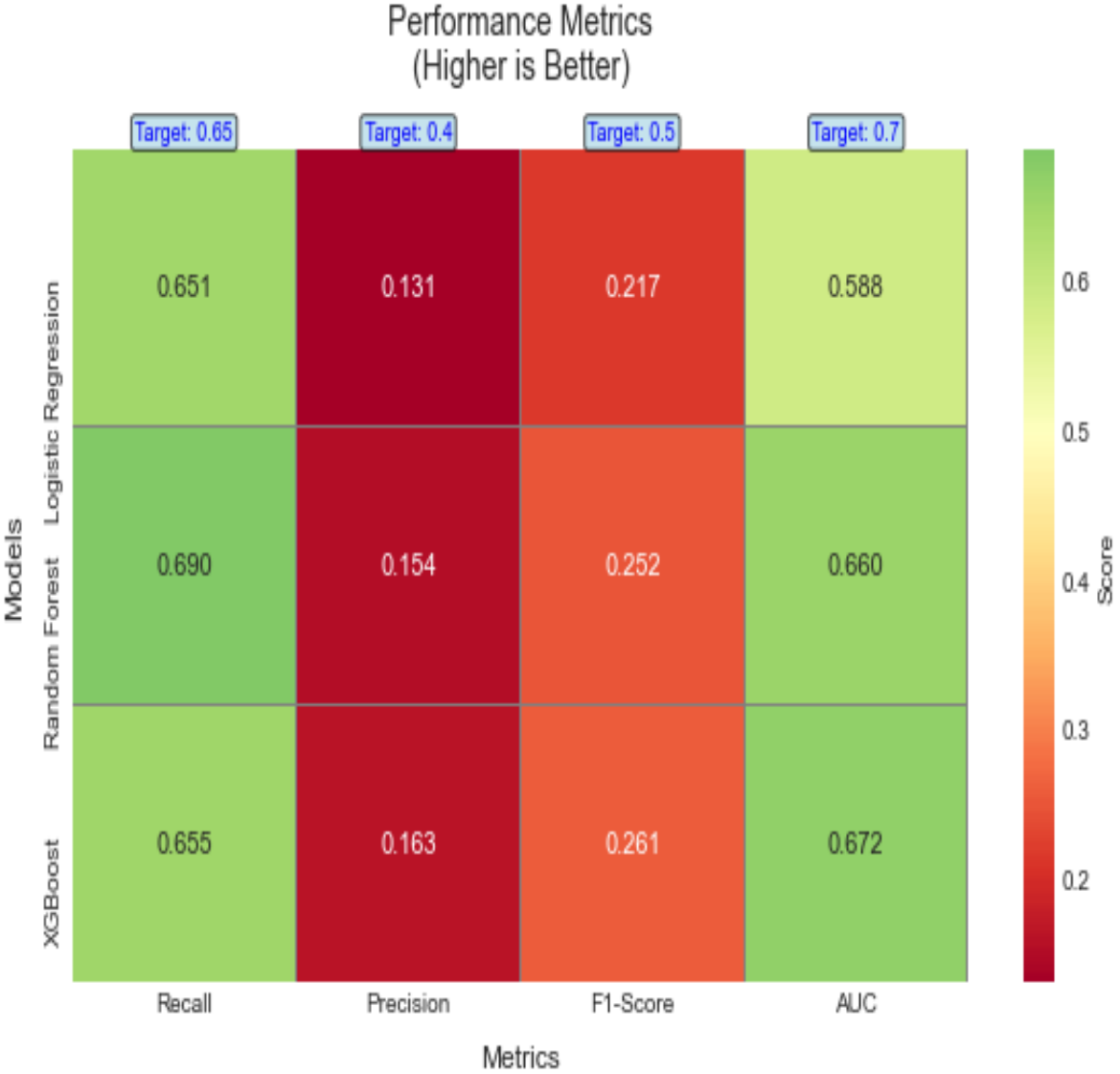
Cost Reduction

Prevent unnecessary readmissions saving an estimated 15-25% of readmission costs through targeted interventions

For a 500-bed hospital, preventing just 10% of readmissions could save \$500,000-\$1M annually



Model Performance Heatmap Comparison

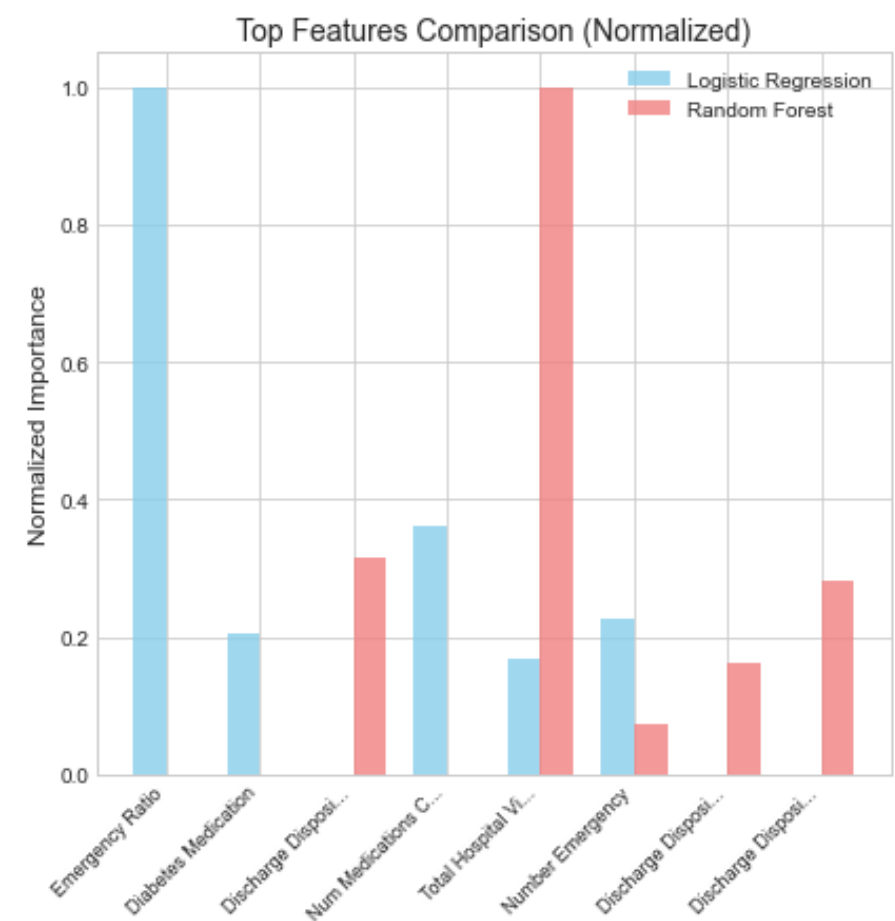
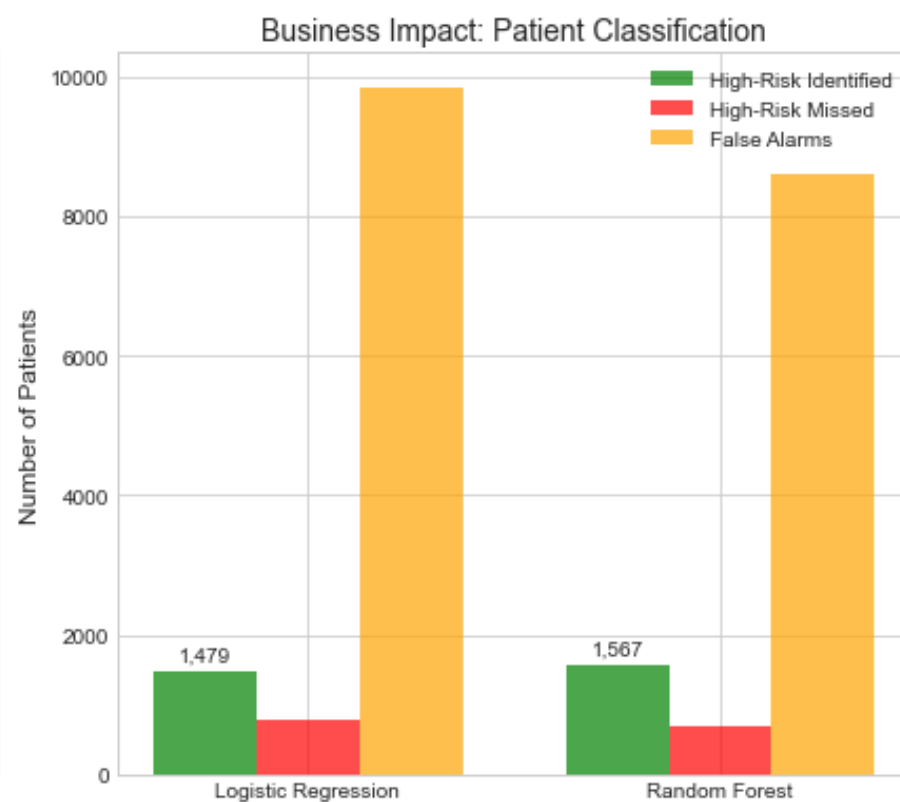
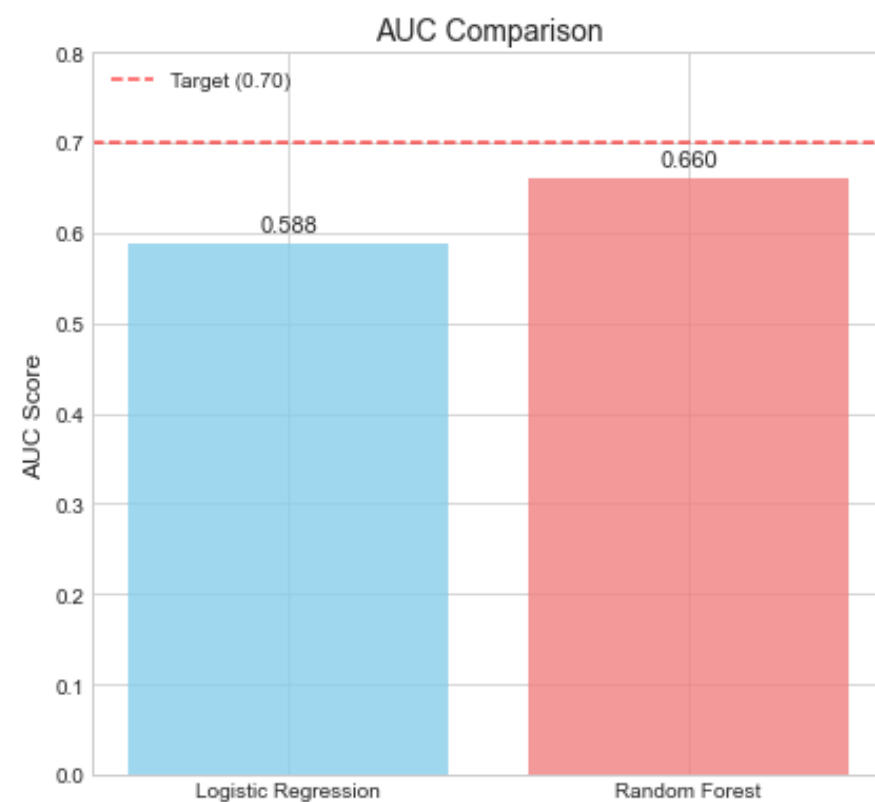
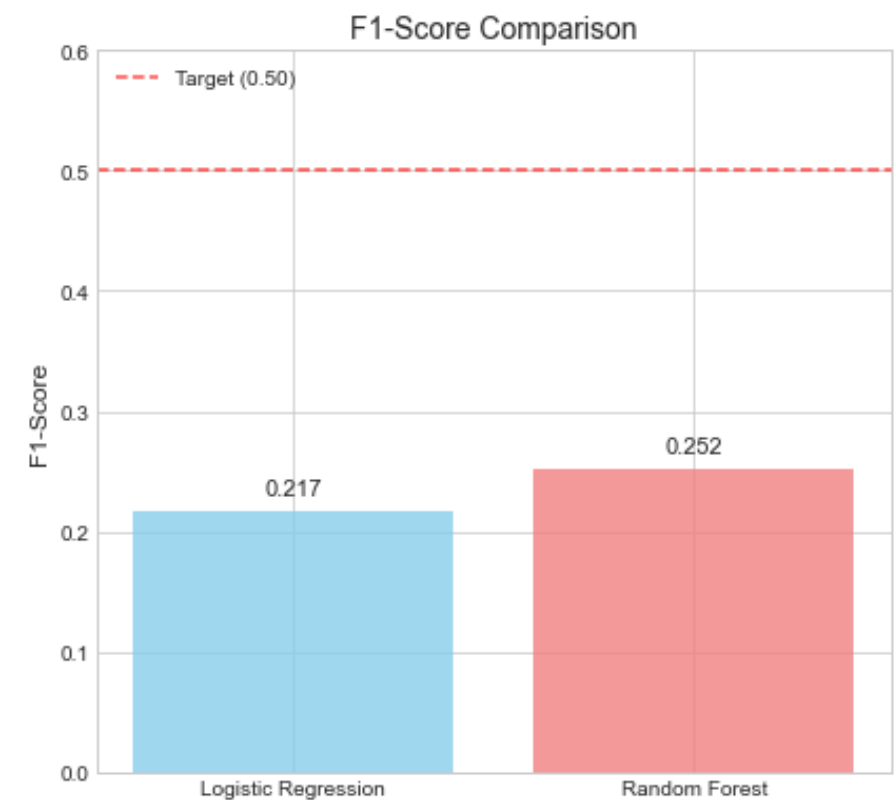
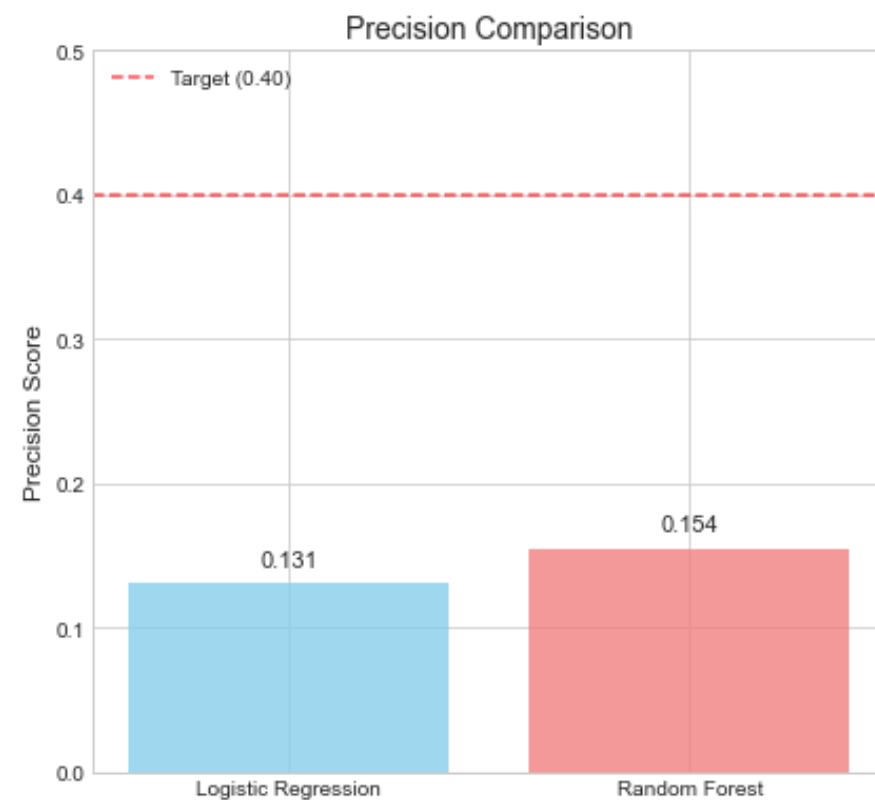
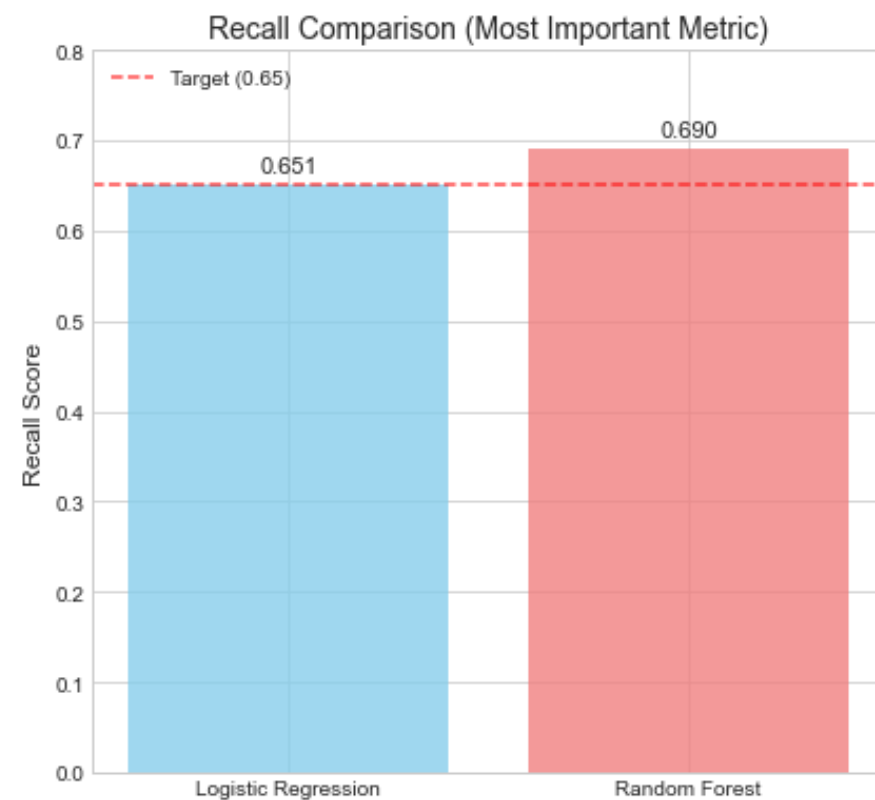


Model Improvement Summary: Logistic Regression

Metric	Baseline	Improved	Target	Status
Recall	0.480	0.651	0.650	☑ MET
Precision	0.159	0.131	0.400	☐
F1-Score	0.238	0.217	0.500	☐
AUC	0.626	0.588	0.700	☐

- Model identifies 65.1% of high-risk patients (Recall target: 65.0%)
- Critical for Kenya: Catches most patients needing intervention
- Trade-off: Low precision means many false alarms (resource consideration)

Model Comparison: Logistic Regression vs Random Forest



Implementation Roadmap

Path Forward

Successful deployment requires collaboration between data science teams, clinicians, and hospital administrators to integrate predictions into existing workflows.



System Integration

Integrate model into electronic health records so risk scores appear automatically at discharge planning.



Clinician Training

Train healthcare providers on interpreting risk scores and implementing appropriate interventions for flagged patients.



Monitor & Refine

Continuously track outcomes and retrain model with new data to improve accuracy over time.



A group of approximately 15 business professionals, both men and women, are standing in a modern office environment. They are all smiling and clapping their hands, suggesting a celebratory or appreciative moment. They are dressed in professional business attire, including suits and blouses. In the foreground, there is a long, dark wooden conference table with several black office chairs on wheels. The office has large glass windows and doors, and the lighting is warm and ambient. The overall atmosphere is positive and professional.

Thank You

Questions and Discussion - We welcome your feedback and insights